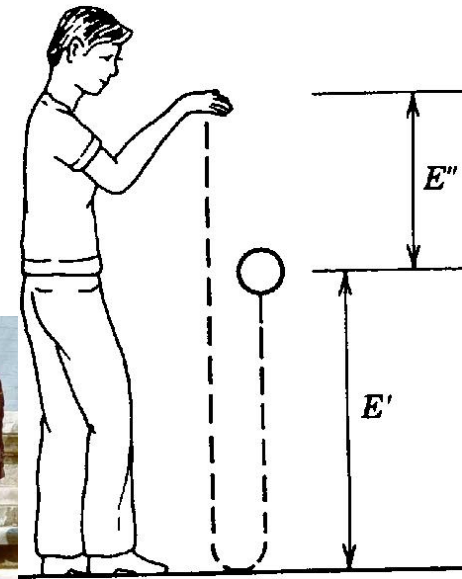
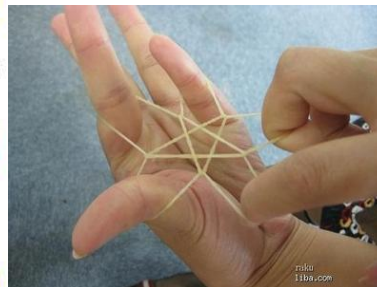


Introduction to Polymer Materials

Lecture 14: Elastomers, Thermosets and their Processing

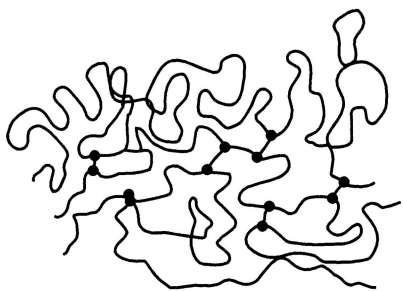
SUN Dazhi



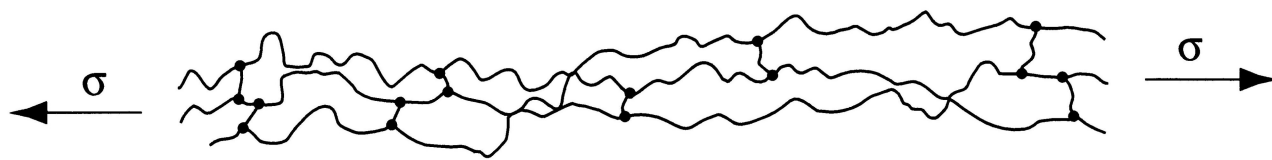
Elastomers

Elastomers are Rubbers ($E < 1$ GPa)

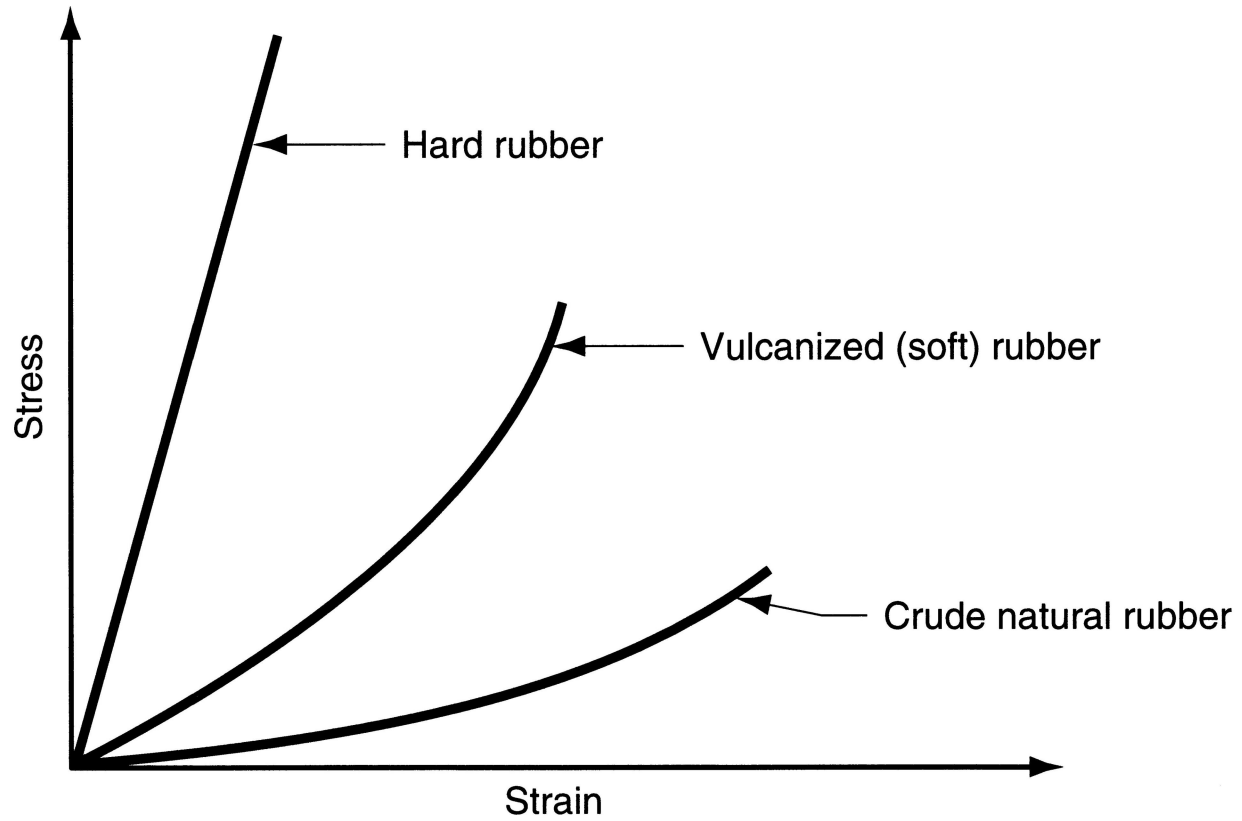
- **Polymers**
- **Amorphous (at least at low strains?)**
- **Very large elongation**
- **T_g : below the operating temperature**
- **Flexibility: low secondary forces**
- **Elastomeric network: moderate degree of crosslinking (Thermosets vs. Thermoplastics)**



(a)



(b)



Common Elastomers/Rubbers

TABLE 16.4 Tabulation of Important Characteristics and Typical Applications for Five Commercial Elastomers

<i>Chemical Type</i>	<i>Trade (Common) Name</i>	<i>Elongation (%)</i>	<i>Useful Temperature Range [°C (°F)]</i>	<i>Major Application Characteristics</i>	<i>Typical Applications</i>
Natural polyisoprene	Natural Rubber (NR)	500–700	–55 to 120 (–65 to 250)	Excellent physical properties; good resistance to cutting, gouging, and abrasion; low heat, ozone, and oil resistance; good electrical properties	Pneumatic tires and tubes; heels and soles; gaskets
Styrene–butadiene copolymer	GRS, Buna S (SBR)	450–500	–60 to 120 (–75 to 250)	Good physical properties; excellent abrasion resistance; not oil, ozone, or weather resistant; electrical properties good, but not outstanding	Same as natural rubber
Acrylonitrile–butadiene copolymer	Buna A, Nitrile (NBR)	400–600	–50 to 150 (–60 to 300)	Excellent resistance to vegetable, animal, and petroleum oils; poor low-temperature properties; electrical properties not outstanding	Gasoline, chemical, and oil hose; seals and O-rings; heels and soles
Chloroprene	Neoprene (CR)	100–800	–50 to 105 (–60 to 225)	Excellent ozone, heat, and weathering resistance; good oil resistance; excellent flame resistance; not as good in electrical applications as natural rubber	Wire and cable; chem. tank linings; belts, hoses, seals, and gaskets
Polysiloxane	Silicone (SIL)	600	–90 to 250 (–130 to 480)	Excellent resistance to high and low temperatures; low strength; excellent electrical properties	High- and low-temperature insulation; seals, diaphragms; tubing for food and medical uses

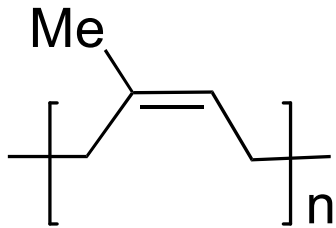
Rubbers

- World usage is ~ 15 million tons
 - Natural rubber is 35%
 - Synthetic rubber is 65%
- **Natural rubber**
 - 75% goes to tires,
 - 5% automotive mechanical parts,
 - 10% non-automotive mechanical parts,
 - 10% miscellaneous parts (medical and health related).

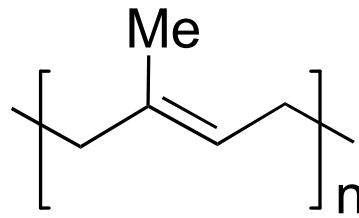


Natural Rubber

- Raw material extracted from trees (Latex)
- Poly-cis-isoprene (~40%) in water



cis



trans



- cis polyisoprene

$T_m = 28^\circ \text{ C}$, $T_g = -70^\circ \text{ C}$

- *trans* polyisoprene (gutta percha)

$T_m = 68^\circ \text{ C}$, $T_g = -70^\circ \text{ C}$

Natural Rubber Processing

Latex is purified, coagulated, and made into sheets or blocks



Natural Rubber Processing

then dried, sorted and smoked



Natural Rubber

Natural rubber in unfilled form

- very large elastic deformations
- very high resilience,
- resistance to cold flow
- resistance to abrasion, wear, and fatigue.

Natural rubber does not have good intrinsic resistance to sunlight, oxygen, ozone, heat aging, oils, or fuels (reactive double bond).

Vulcanizes with 4% sulfur

Rubber Additives and Modifiers

Fillers can comprise half of the volume of the rubber

- Silica, carbon black, metal oxides,
- Reduce cost of material.
- Increase tensile strength and modulus.
- Improve abrasion resistance.
- Improve tear resistance.
- Improve resistance to light and weathering.

Example:

- Tires produced from Latex contains 30% carbon black which improves the body and abrasion resistance in tires.

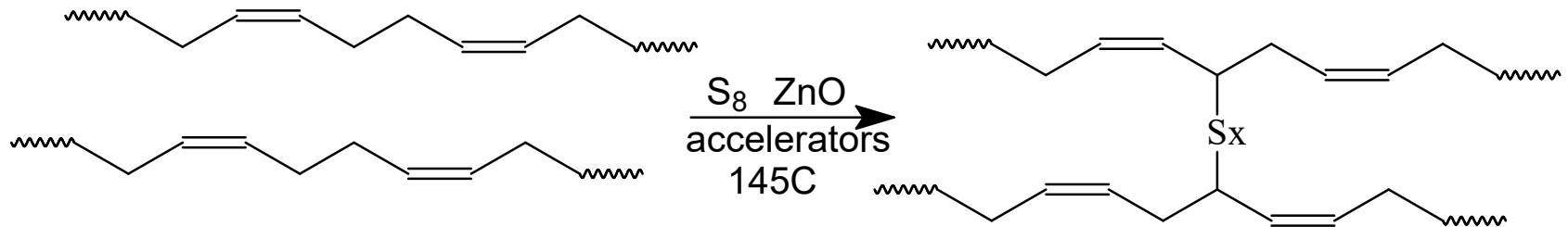
Rubber Additives and Modifiers

Additives

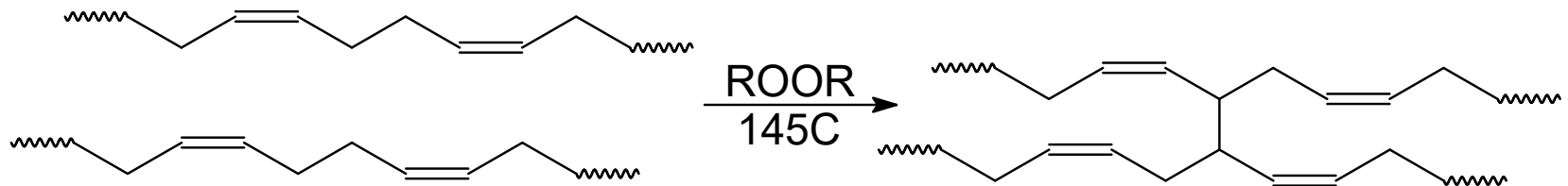
- Antioxidants,
- Antiozonants
- Light stabilizers/UV absorbers
- Anti-aging agents
- Crack protecting agents
- Anti-fogging agents
-

Vulcanization - Sulfur and Peroxide Chemistry

- Sulfur Cures: applied only to unsaturated materials



- Peroxide Cures: can be used on most every polymer



Vulcanization, **curing** and **crosslinking** are equivalent terms referring to the process by which individual polymer chains are transformed into a network.

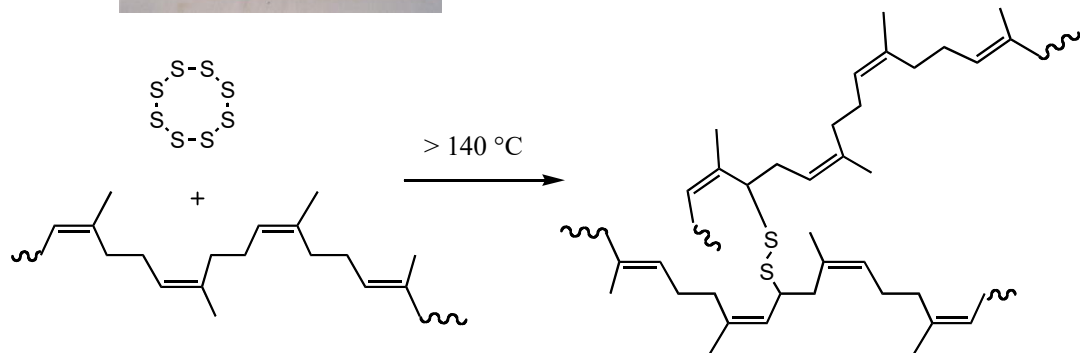
Goodyear Rubber



Charles Goodyear
(1800 - 1860)

**Invented vulcanization
of rubber in 1839**

Video



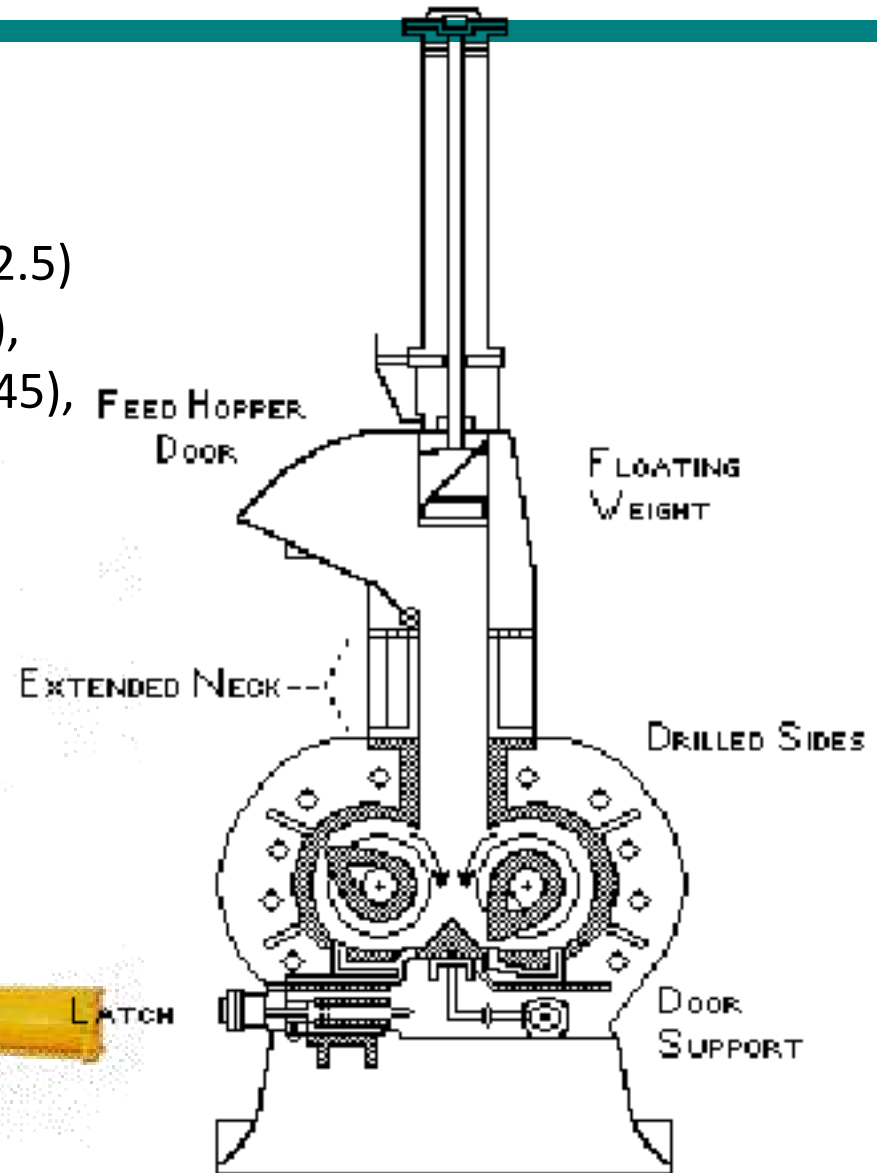
Enabled commercialization of natural rubber

Rubber Processing

Compounding

Typical tire:

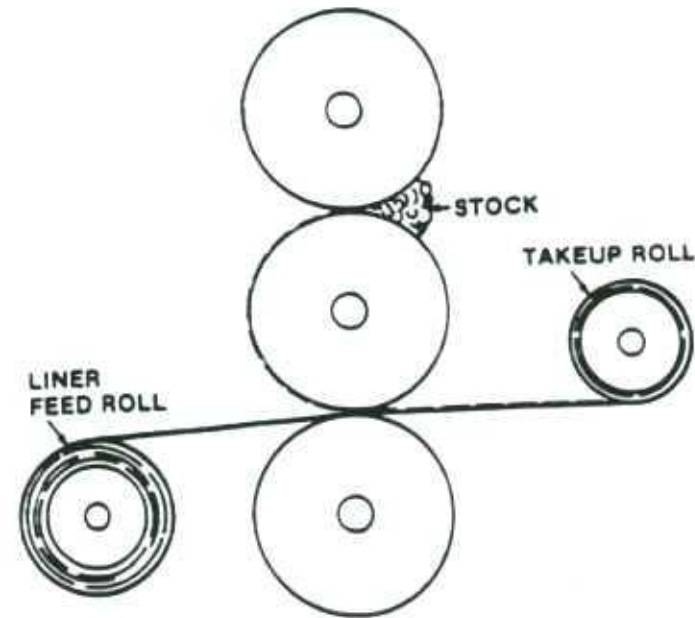
Natural rubber smoked sheet (100), sulfur (2.5)
sulfenamide (0.5), MBTS (0.1), steric acid (3),
zinc oxide (3), PNBA (2), HAF **carbon black** (45),
and mineral oil (3)



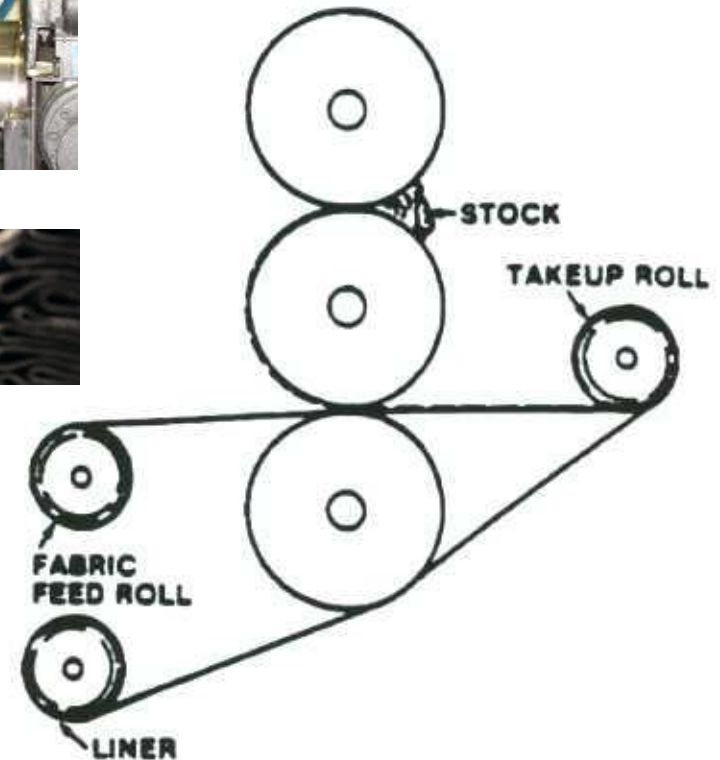
Calendering

Calendering is the process for producing long runs of uniform thickness sheets of rubber either unsupported or on a fabric backing.

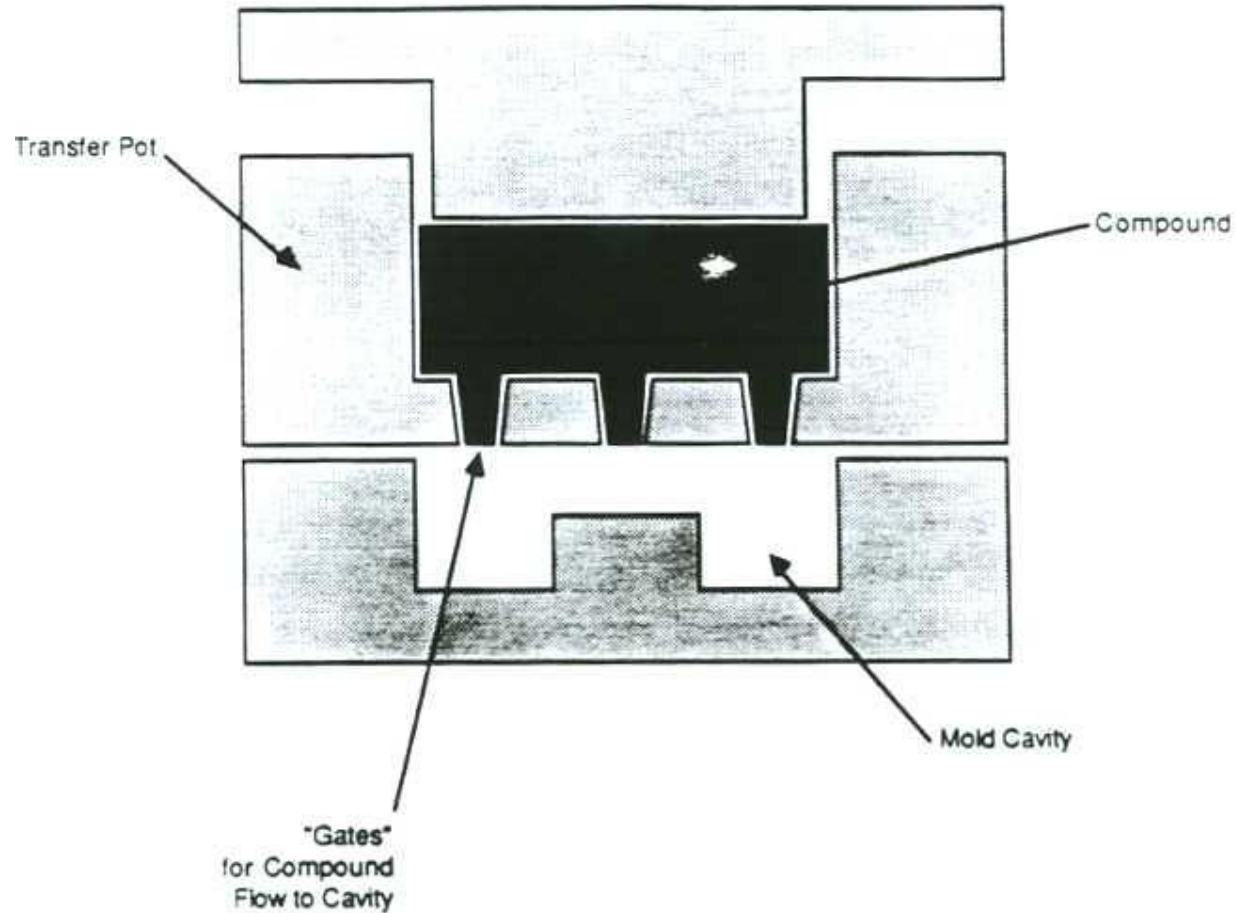
Unsupported Silicone Sheet



Supported Silicone Sheet



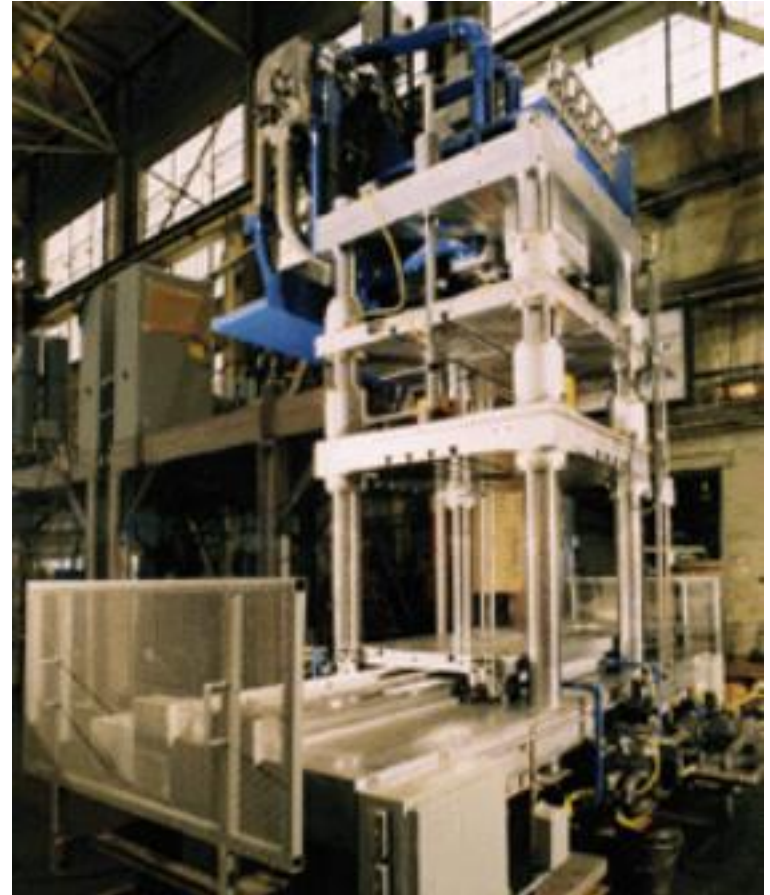
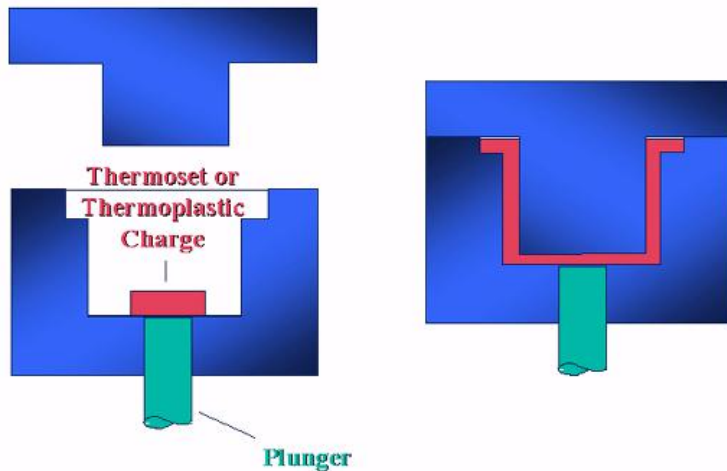
Transfer Molding



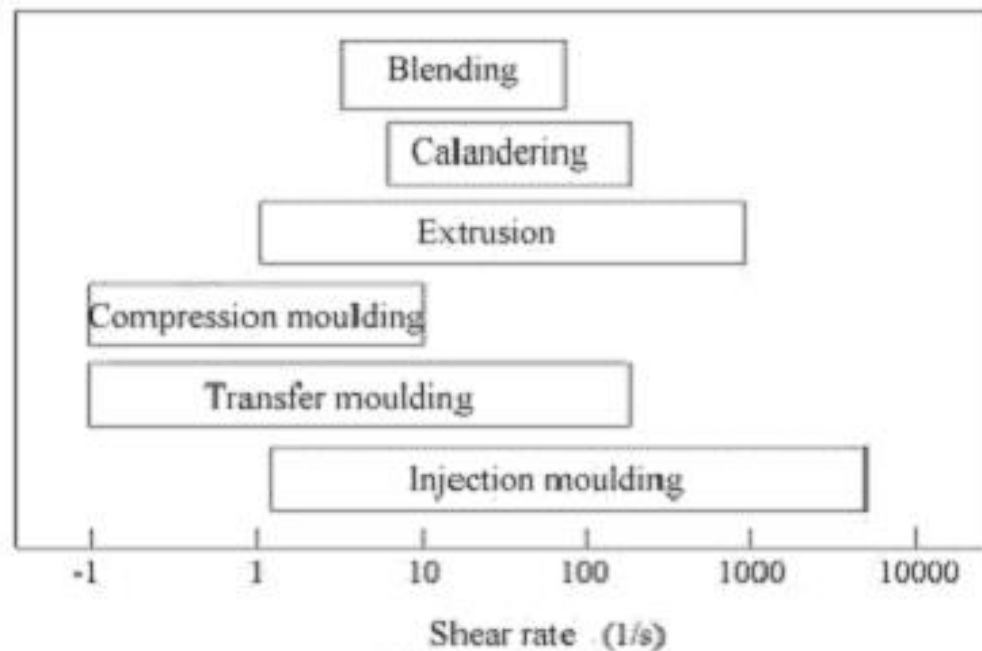
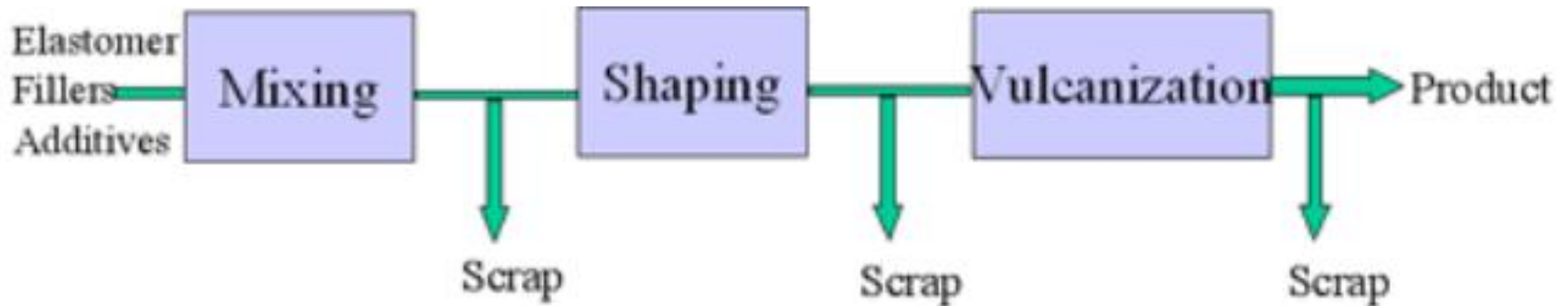
Transfer Molding

Compression Molding Process

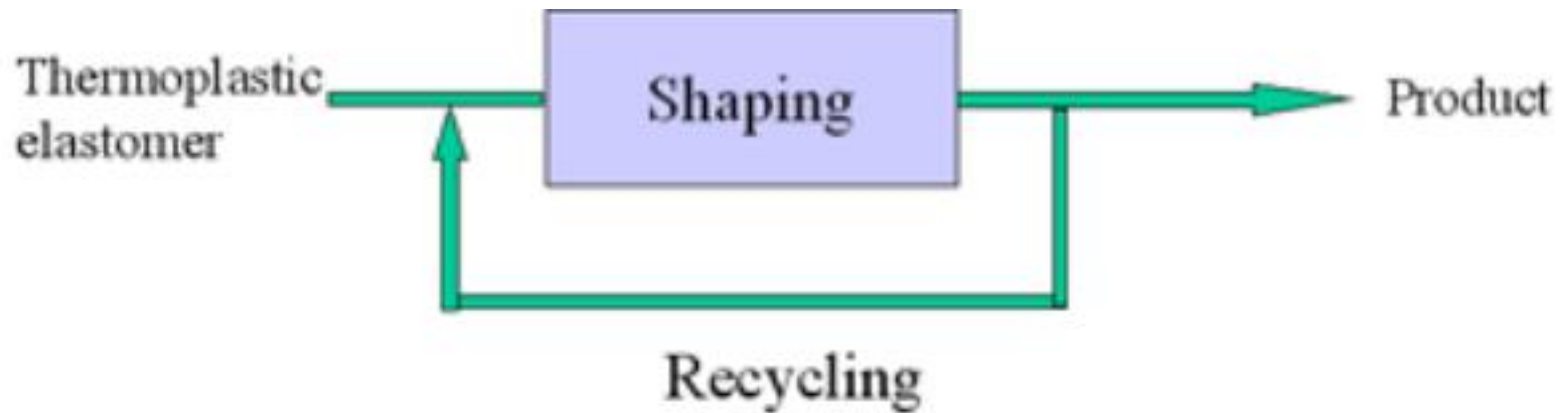
Compression Molding Process



Elastomer Processing

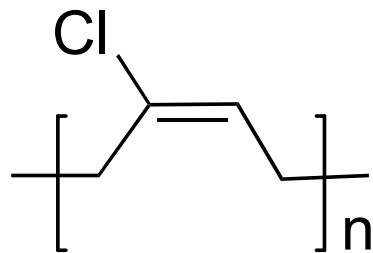


Elastomer Processing



Polychloroprene

- Polychloroprene or **Neoprene** was the very first synthetic rubber (DuPont 1930, commercialized in 1937)
- Due to polar nature of molecule from Cl atom it has very good resistance to oils (**Oil-Resistant Elastomers**) and is flame resistant (Cl gas coats surface)
- Used for fuel lines, hoses, gaskets, cable covers, protective boots, bridge pads, roofing materials, fabric coatings, and adhesives

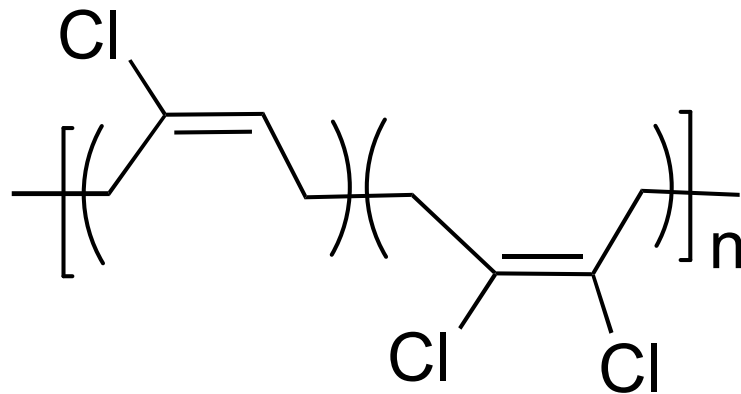


polychloroprene
or Neoprene

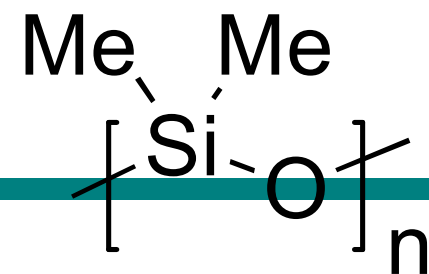


Polychloroprene

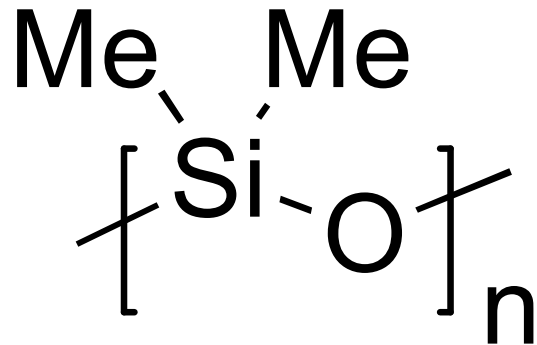
- $T_g = -65^{\circ} \text{ C}$
- Slowly crystallizes & hardens below 10° C
- Copolymer with 2,3-dichlorobutadiene won't crystallize



Silicones



Silicone Rubbers

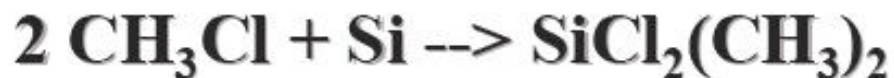
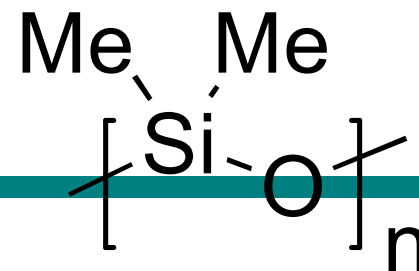


Polydimethylsiloxane

$$T_g = -123^{\circ} \text{C}$$

- Si-O replaces C-C backbone
- Chemically Inert
- Low conductivity
- Heat/cold resistant
- Relatively expensive
- X-linking increases stiffness and strength.

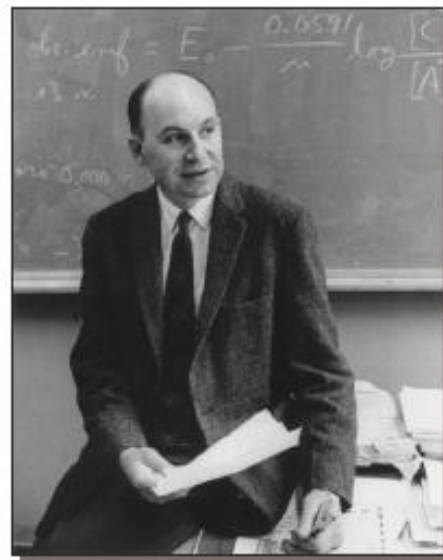
Silicones



Also produces $\text{SiCl}_3(\text{CH}_3)$,

$\text{SiCl}(\text{CH}_3)_3$, and SiCl_4

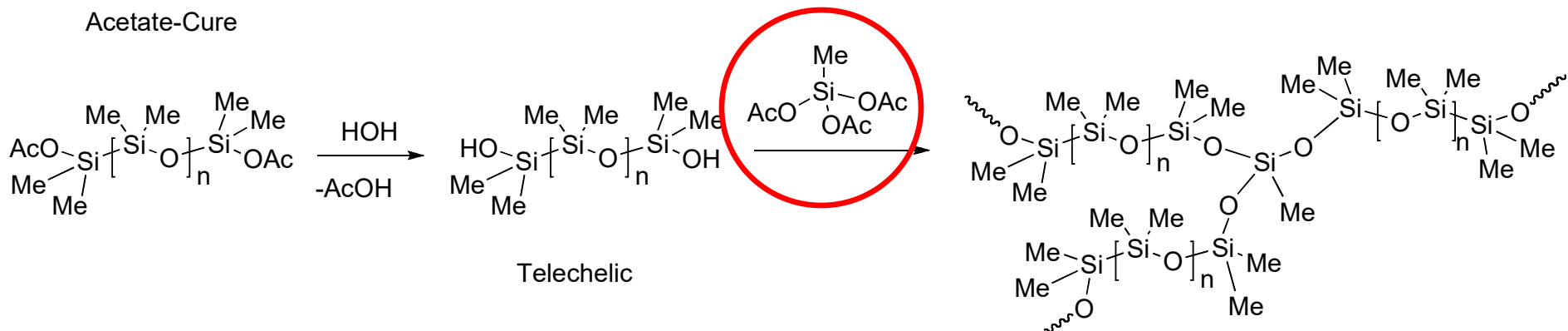
Patented by E. G. Rochow of GE
in 1945



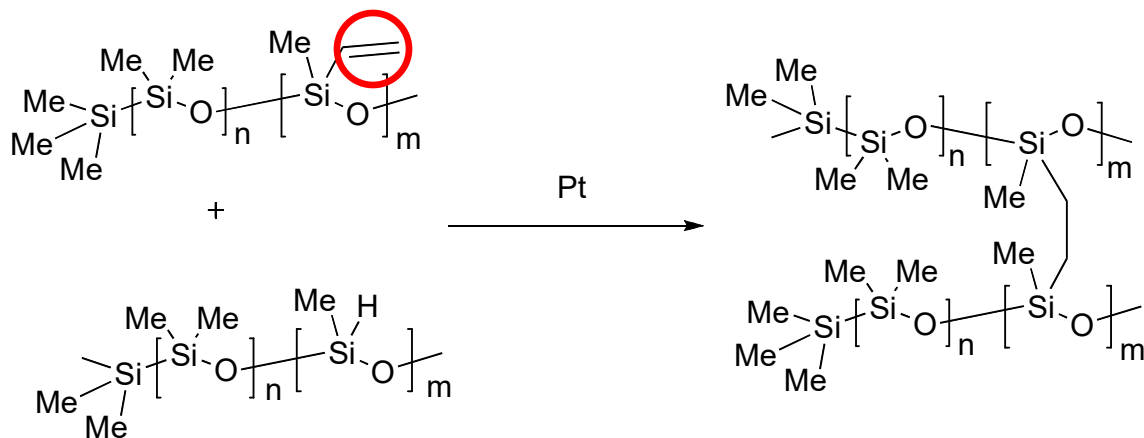
Eugene Rochow

Vulcanization of Silicones

Acetate-Cure



Hydrosilation or Platinum cure



Thermoplastic Elastomers

- Five types
 - Olefinics
 - Fluoropolymers
 - Styrenics
 - Polyurethanes
 - Polyesters
- Use physical cross-links to “vulcanize” the polymer

Thermoplastic Elastomers

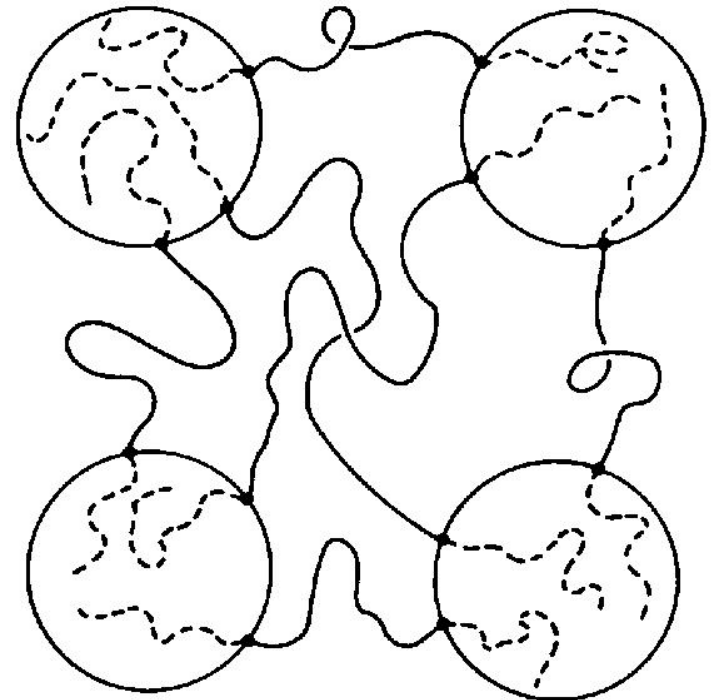
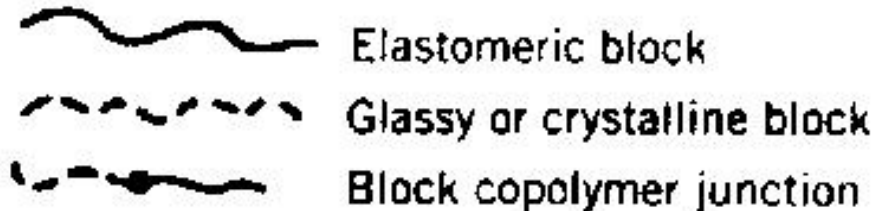
- Processing involves melting of polymers, not thermoset reaction
- Processed by injection molding, extrusion, blow molding, film blowing, or rotational molding.
 - Injection molded soles for footwear



Thermoplastic Elastomers

Di-block (or more) copolymers consisting of a 'soft' elastomeric segment and two 'hard' amorphous blocks.

- Under processing conditions, both segments are above T_g , allowing the material to flow.
- On cooling, separation of the phases into two domain types creates physical crosslinks between molecules.



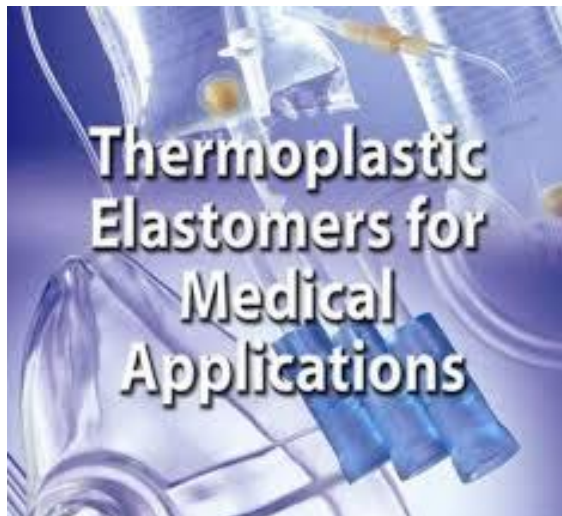
Thermoplastic Elastomers

–Advantages of thermoplastic elastomers

- Less expensive due to fast cycle times
- More complex designs are possible
- Wider range of properties due to copolymerization

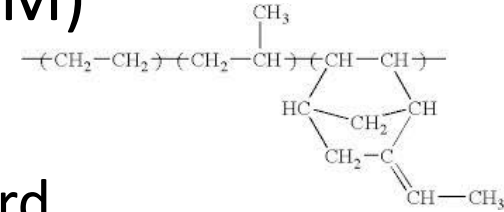
–Disadvantage of thermoplastic elastomers

- Higher creep

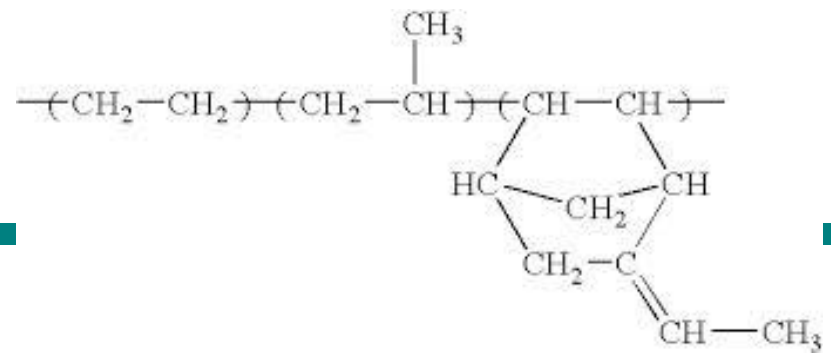


Olefinic Thermoplastic Elastomers

- Ethylene-Propylene-Diene Monomer (EPDM)
- More easily processed
 - Injection molding, extrusion and other standard thermoplastic processes
 - Highly compatible with polyolefins
 - EPDM is crosslinked very lightly and may not be capable of being melted
- 7-21 MPa Ultimate Tensile
- Service range: -50 ° C-150 ° C
- 100-600% elongation



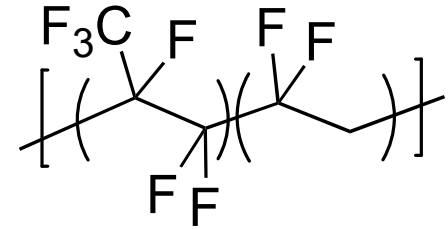
EPDM



<u>Characteristics</u>	<u>High</u>	<u>Low</u>
Ethylene Content	Good Green Strength Flow at High Extrusion Temperatures High Tensile Strength, Modulus High Loading (Reduced Cost)	Fast Mixing Low Temperature Flexibility Low Hardness and Modulus Calendering and Milling
Diene Content	Cure Degree and Fast Rate Acceleration Versatility Good Compression Set High Modulus, Low Set	Scorch Resistance High Heat Stability Low Hardness and Modulus
Molecular Weight	Good Tensile, Tear, Modulus, Set High Loading and Oil Extension Good Green Strength Collapse Resistance	Fast Mixing High Extrusion Rates Good Calendering Low Viscosity, Scorch Resistance
MWD	Overall Good Processing Extrusion Feed and Smoothness Collapse Resistance Good Calendering and Milling	Low Die Swell Fast Extrusion Rate High Cure Good Physicals

Fluoropolymer elastomers

- Terpolymers
- Viton, Dynecon, Aflas, Kalrez, Chemraz
- most chemically resistant of all elastomers
 - resistant to acids, caustics, amines, aldehydes, steam, and salt water
- very expensive
- Only available as o-rings and sheets
- Amorphous



Hexafluoropropylene-vinylidene fluoride copolymer

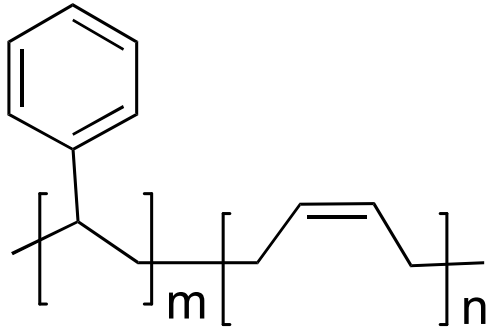
- Use range: -40 to 200°C
- Excellent resistance to petroleum products and solvents.
- Very good high-temperature performance.
- Fluorocarbon elastomers make up the most widely used seals in the semiconductor industry.

Tensile Strength 12.1 MPa, Elongation 210%



Styrene Butadiene Rubber (SBR)

Styroflex[®] SBS (Styrene-Butadiene Copolymer)



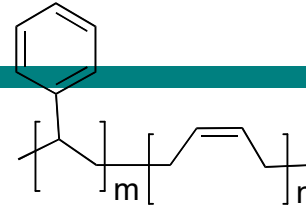
Styrene-Butadiene Block Copolymer (or **SBS**)

Tensile Strength = 3 MPa

Tensile Modulus = 130 MPa

Elongation at break 550%

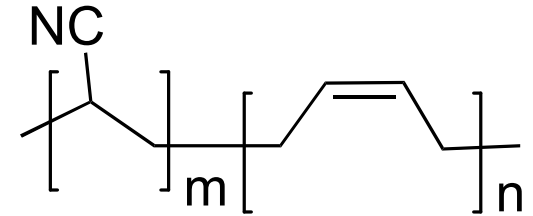
Styrene Butadiene Rubber (SBR)



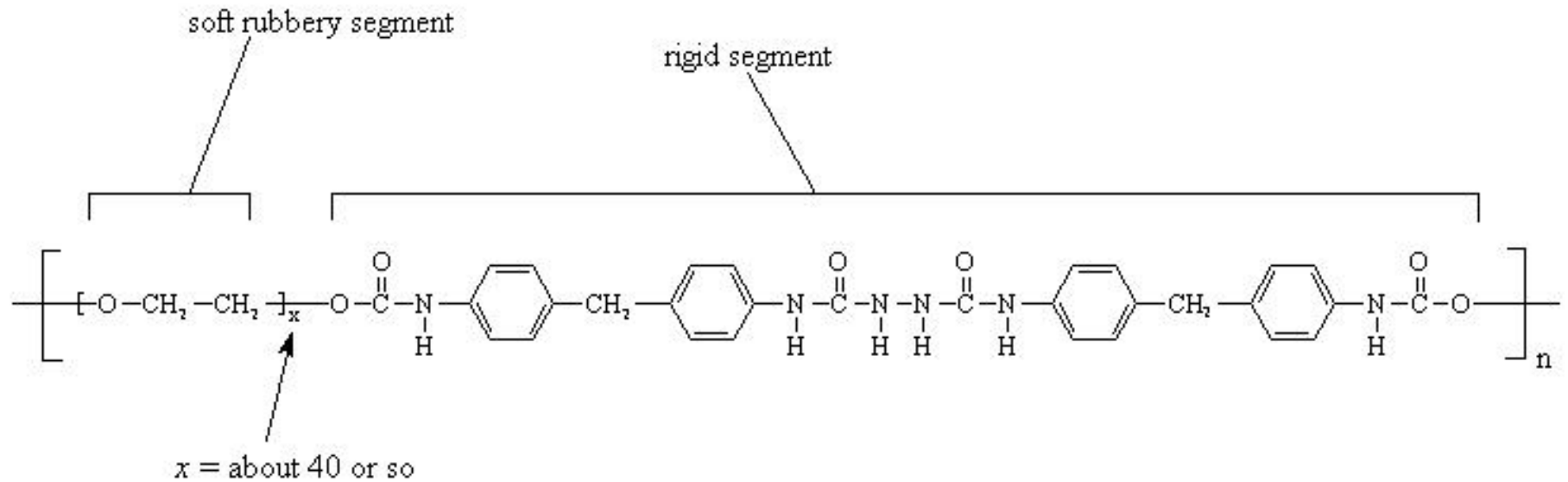
- Developed during WWII
- Random copolymer of butadiene (67-85%) and styrene (15-33%)
- Tg of typical 75/25 blend is -60°C
- Requires reinforcing filler, carbon black, to get good properties.
- One of the least expensive rubbers and generally processes easily.
- Inferior to natural rubber in mechanical properties
- Superior to natural rubber in wear, heat aging, ozone resistance, and resistance to oils.
- Applications include tires, footwear, wire, cable insulation, industrial rubber products, adhesives, paints (latex or emulsion)
- ✓ **More than half of the world's synthetic rubber is SBR**
- ✓ **World usage of SBR equals natural rubber**

Nitrile Butadiene Rubber(NBR)

- Copolymerization of butadiene and acrylonitrile
- More expensive than SBR or BR
- Solvent resistant rubber due to nitrile C:::N
(Oil-Resistant Elastomers)
- Irregular chain structure will not crystallize on stretching, like SBR
- vulcanization is achieved with sulfur like SBR and natural rubber

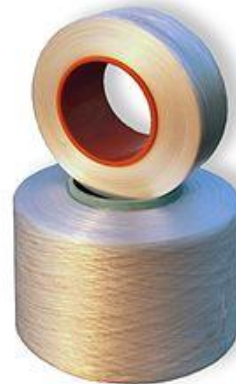


Polyurethane Elastomers: Spandex



Invented by Joseph Shivers in DuPont in 1958.

Spandex has a complicated structure, with both urea and urethane linkages in the backbone chain.



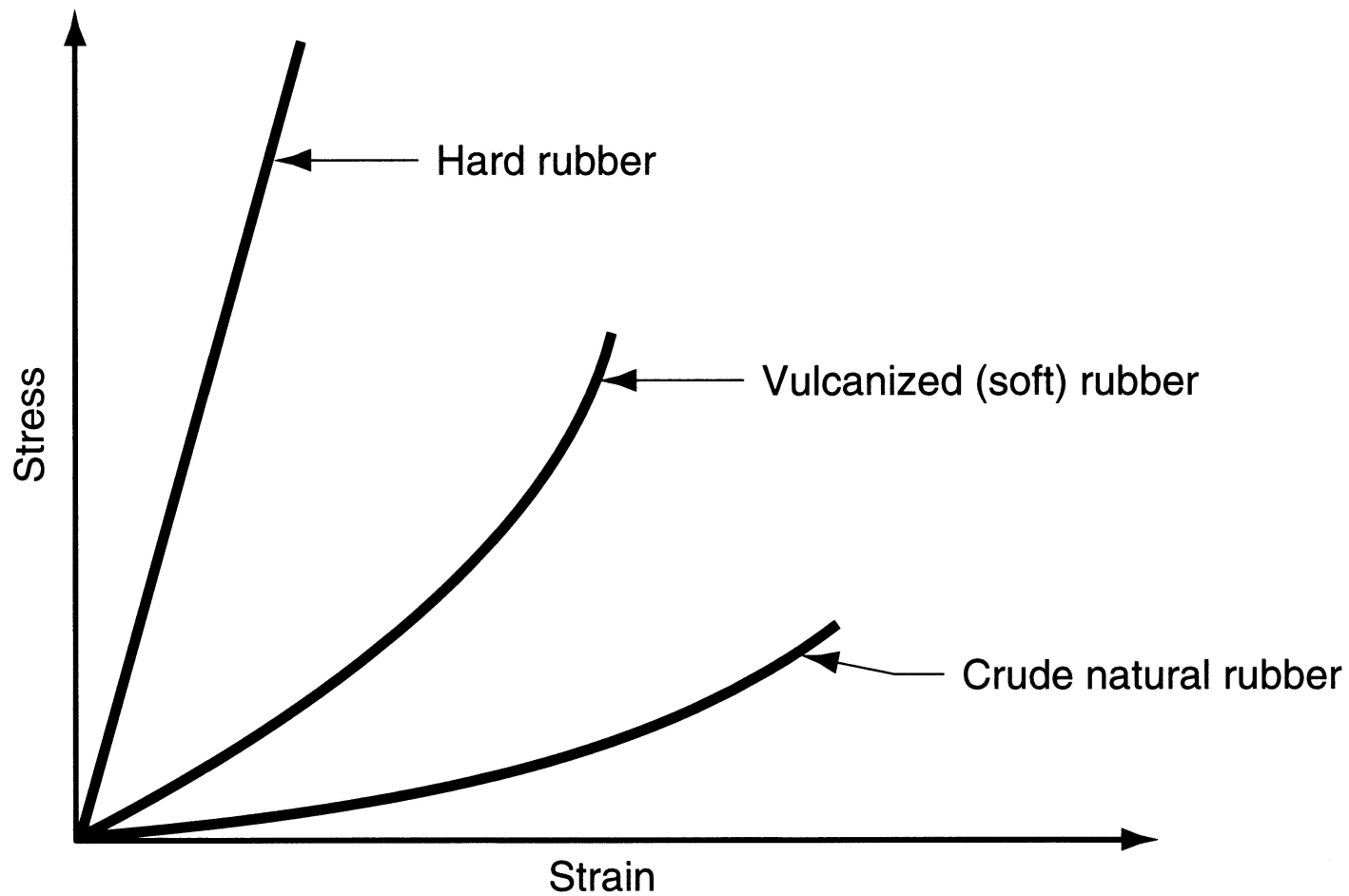
Polyester Thermoplastic Elastomers



Riteflex® MT9000 series of **copolyester** thermoplastic elastomers (TPE) are certified for use in drug delivery systems, medical devices, pharmaceutical and other healthcare applications, as well as in repeat-use, food-contact applications

Thermoset Plastics

- Polymerization by chemical reaction of two or more monomer resins often at high temperature, with catalysts or mixing
- Product is insoluble, often intractable
- Generally not reversible
- Product manufacturing by forming shape during reaction
- Examples and applications:
 - Polyurethanes: Sport boots, convey belt, coatings
 - Phenolic: circuit board, billiard balls, coatings
 - Epoxy: paint, adhesives, composites
 - Unsaturated polyesters
 - Vinyl esters
 - Bismaleimides
 - **Melamines**



Thermosets



Liquid Monomer(s),
Oligomeric
Precursors,
Thermoplastic
with curable groups
Elastomer

Chemical
reaction

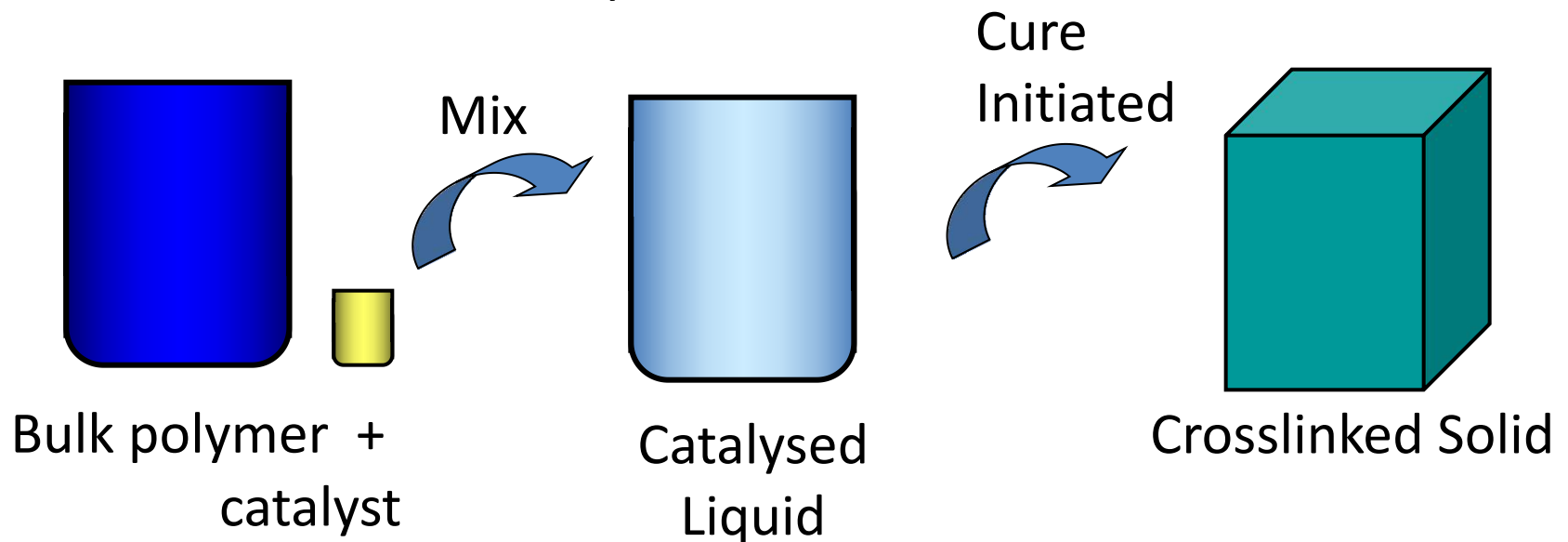
Glassy Thermoset
or
Vulcanized Elastomer

Molding: Complex shapes-vulcanizing elastomers

Reaction Injection Molding: RIM: Mixing two monomers or precursors

Thermoset Processing

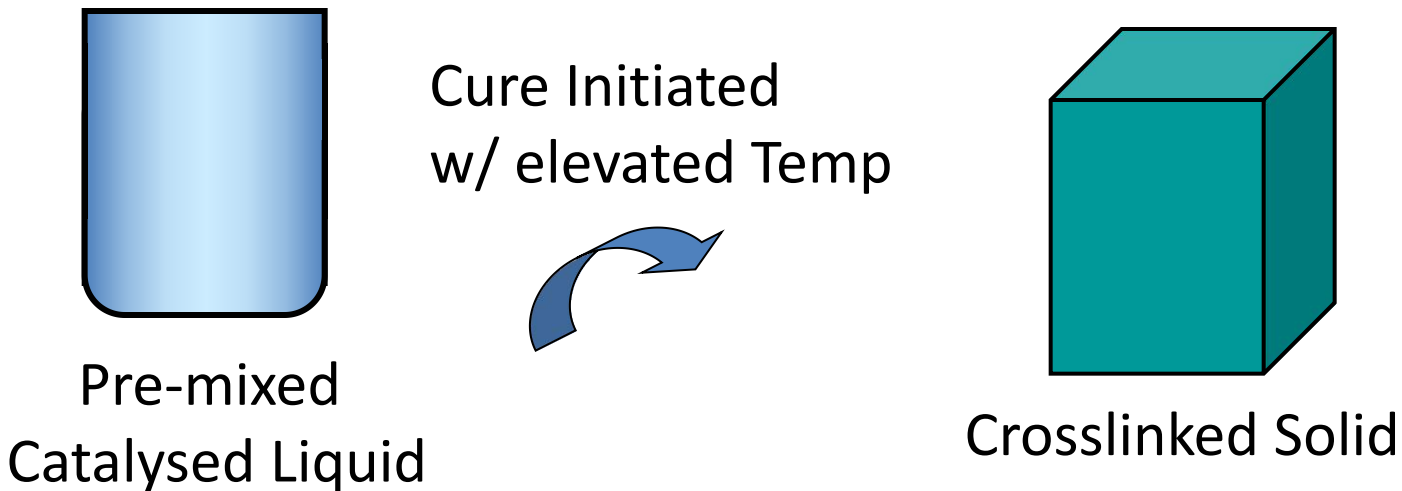
- **Two piece liquids** - Vast bulk of mixture formed by the **bulk resin**), with **catalyst**. Mixing carried out by the user.



No solvent- eliminates waste & shrinkage

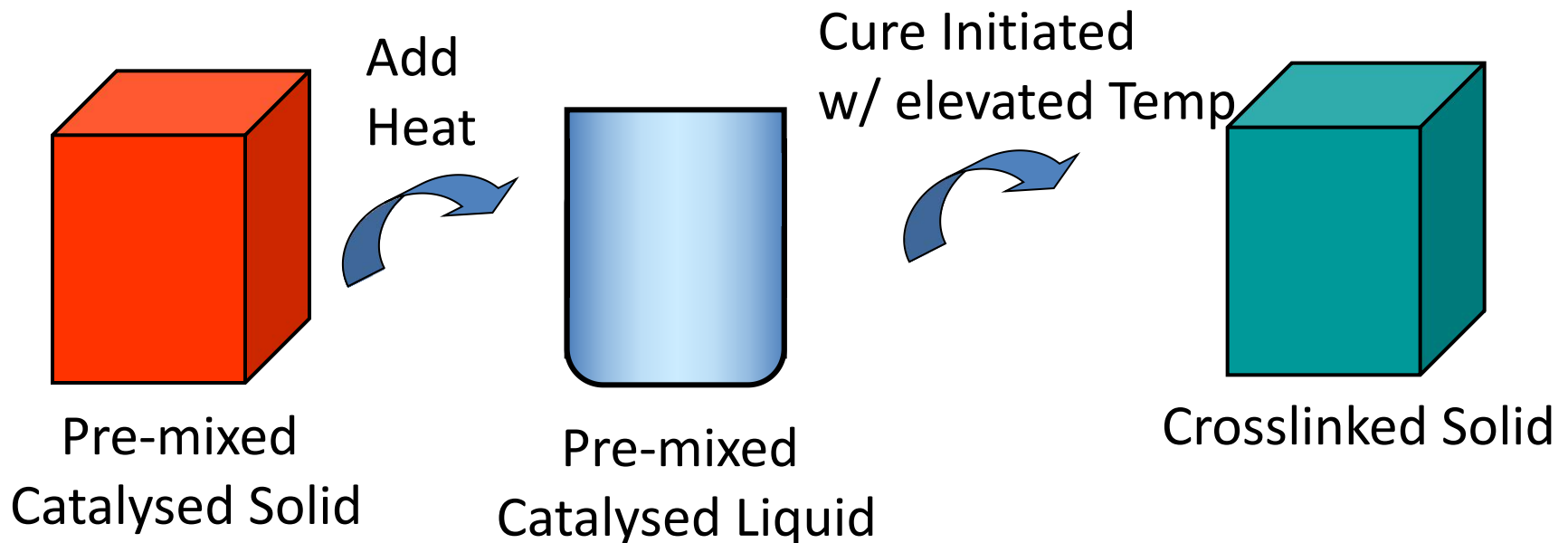
Thermoset Processing

- **One piece liquids** - All components **pre-mixed** by the manufacturer. Cure is initiated by increased temperature, pressure, or exposure to UV light.



Thermoset Processing

- **One piece Solids** - All components pre-mixed by manufacturer, and the solution remains solid at room temperature. Heat is applied to raise the resin above melting temperature. Further heat and pressure are applied to initiate cure.



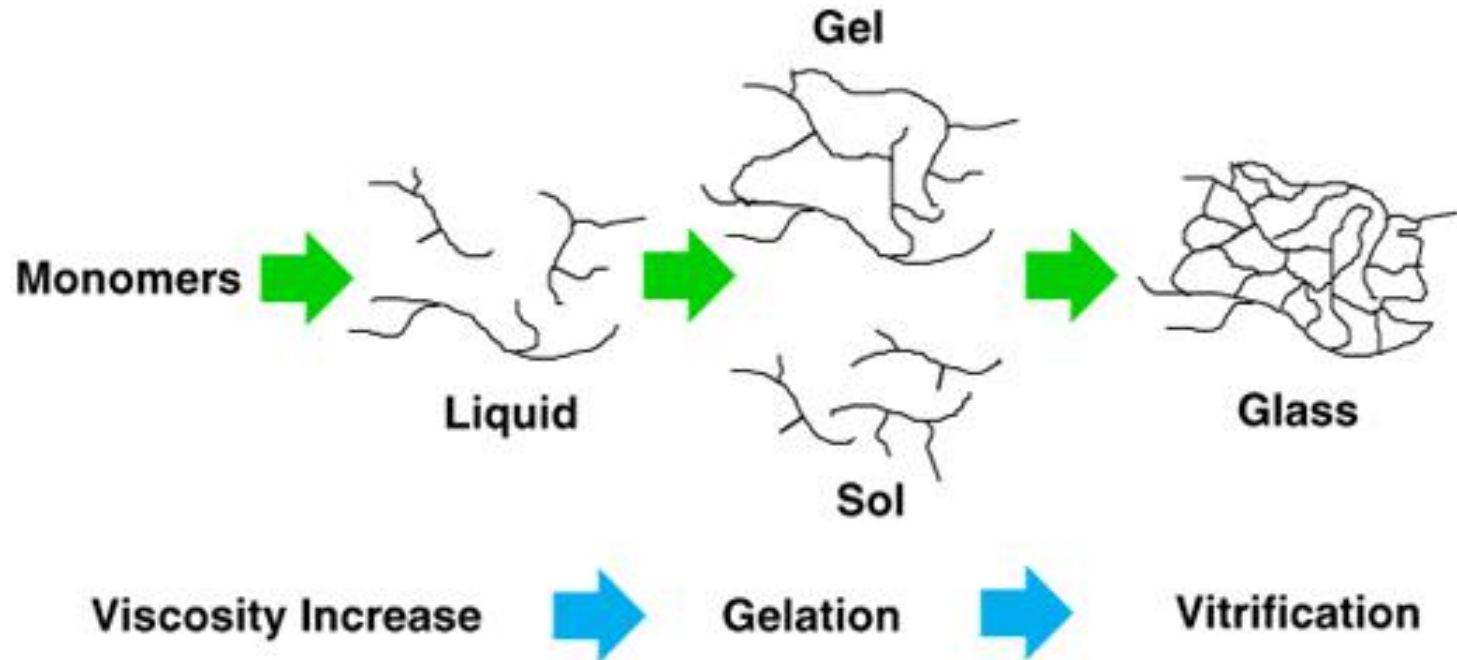
Thermoset Plastics

As compared to TP's, TS's are brittle at room temperature, and cannot be reshaped due to the strong cross-linking covalent bonds.

However, their comparative advantages include;

- Higher tensile strength and stiffness
- Excellent chemical and solvent resistance
- Good dimensional and thermal stability
- Good creep resistance
- Excellent fatigue properties
- Low viscosities, simplifying physical processing

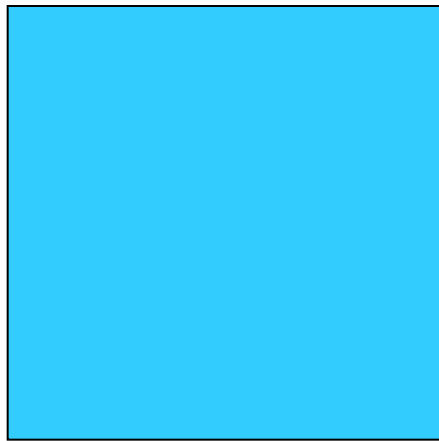
Gelation and Vitrification



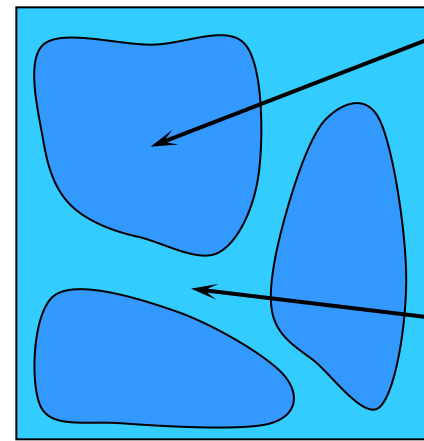
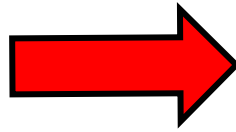
- Gelation:**
- Point at which covalent bonds begin to connect between linear chains, forming regions of large networks.
 - The resin transforms from a liquid to a rubbery state.
 - The reaction continues at a significant rate.
 - There is a drastic increase in viscosity.

Gelation

at Gelation:



Liquid



Crosslinked
networks

Catalysed
solution

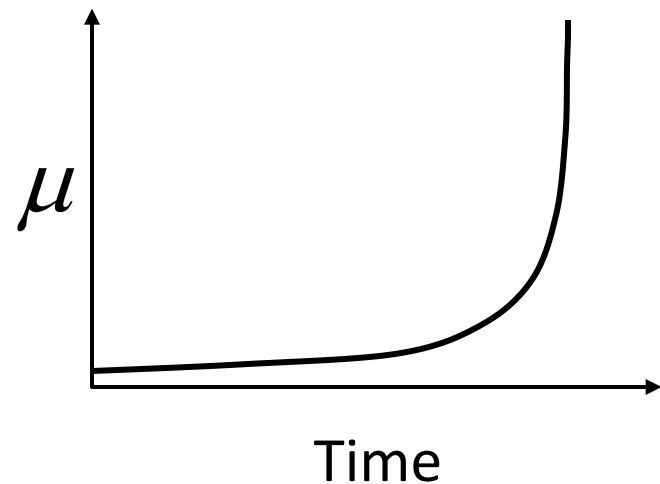
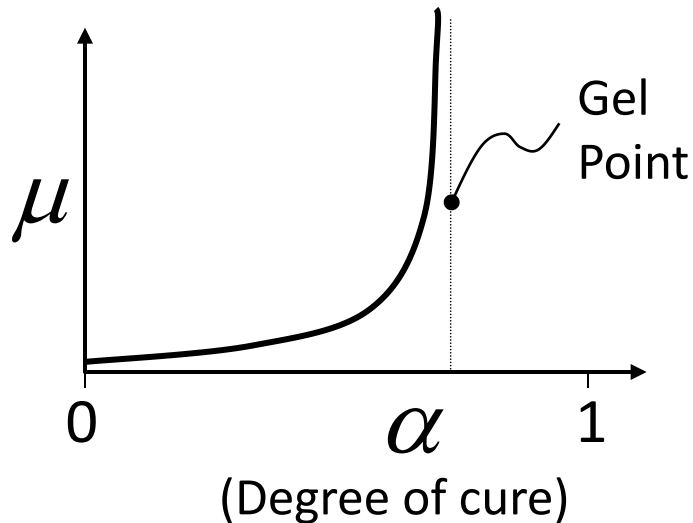
Gel: A two-phase structure.

Vitrification

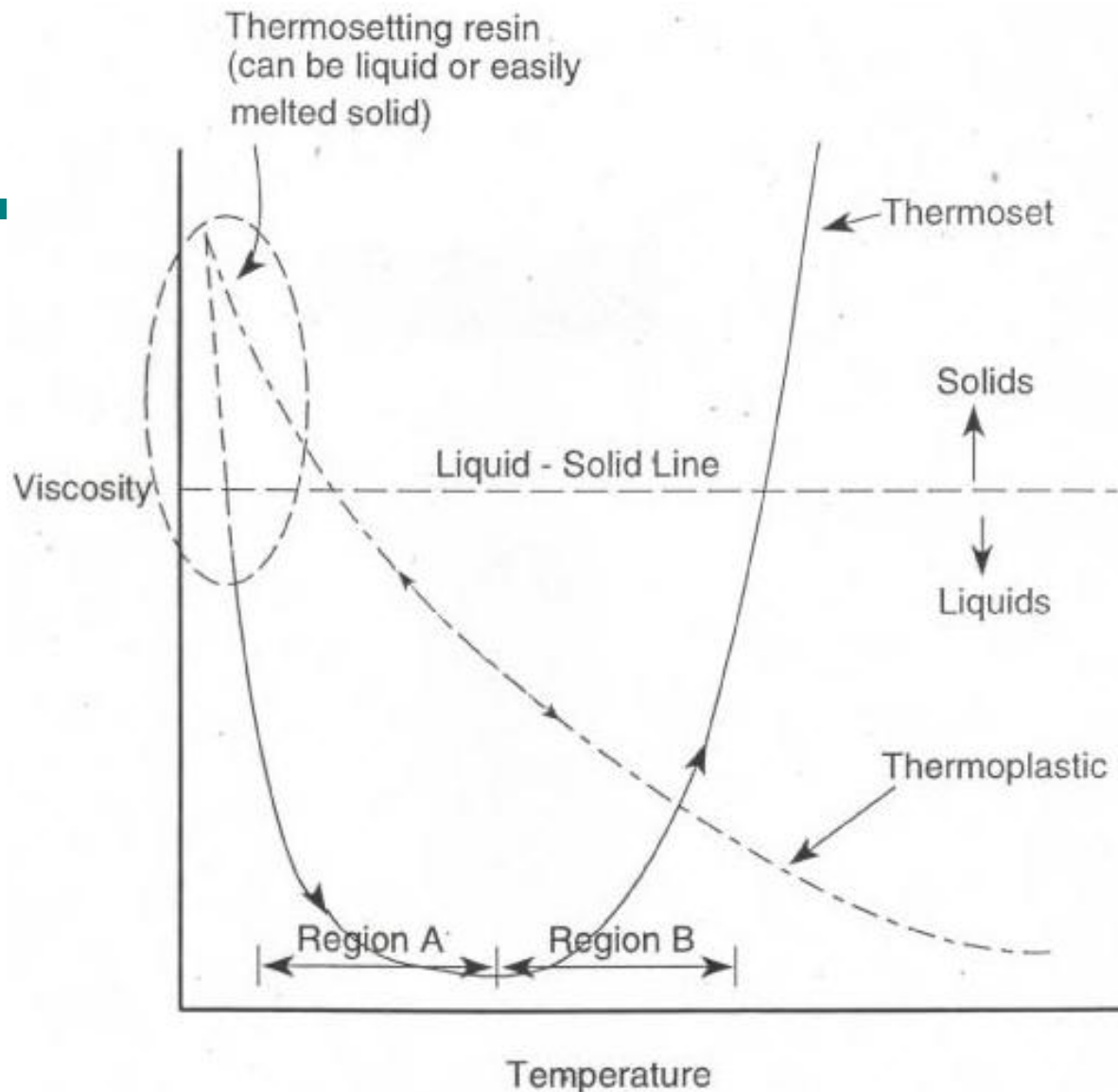
- Occurs when the glass transition temperature of the curing resin increases to the current resin temperature.
- The rate of the cure reaction is significantly reduced, as further crosslinking requires diffusion of molecules through the network.
- The final physical phase depends on the temperature the process has been held at.

Viscosity

During the early stages of cure, the pre-polymeric chains are combining, and the average molecular weight of the mixture increasing. The viscosity therefore increases.



Initially the increase will be relatively slow, significantly quickening as the “**gel point**” is reached.

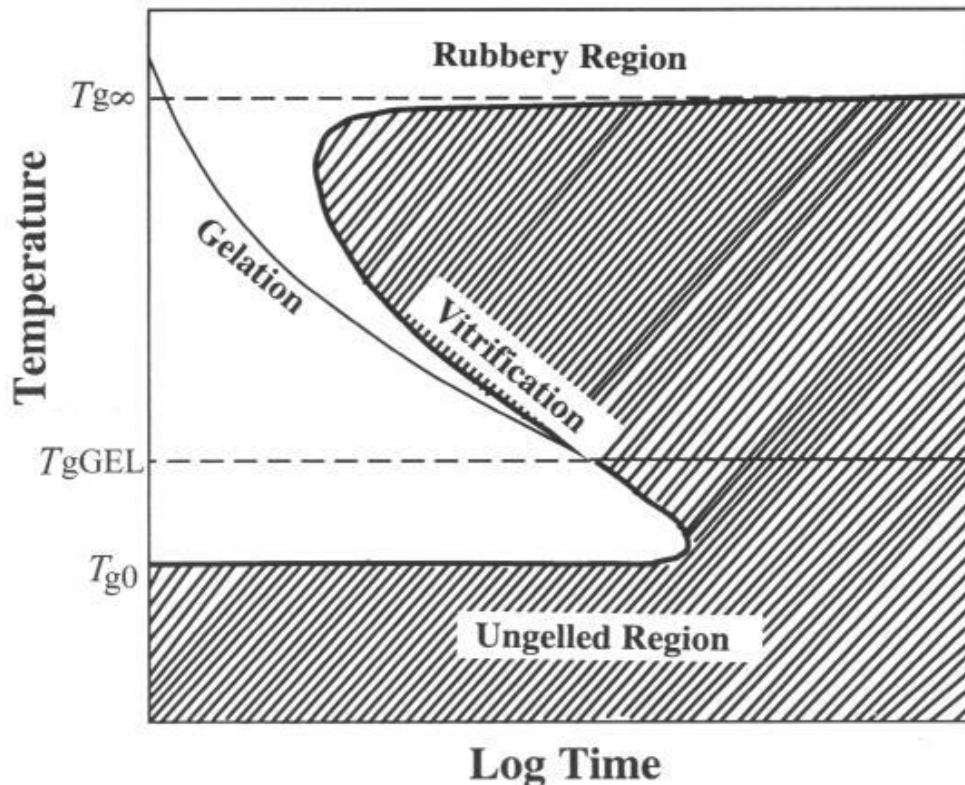


Region A is dominated by thinning from temperature increases
Region B is dominated by thickening from crosslinking

Phase Transformations

Thermoset resins may assume various physical forms, or phases during the cure reaction, depending on the temperature history.

The influence of **temperature** and **time** is best appreciated through consultation of the **Time-Temperature-Transformation (TTT)** diagram

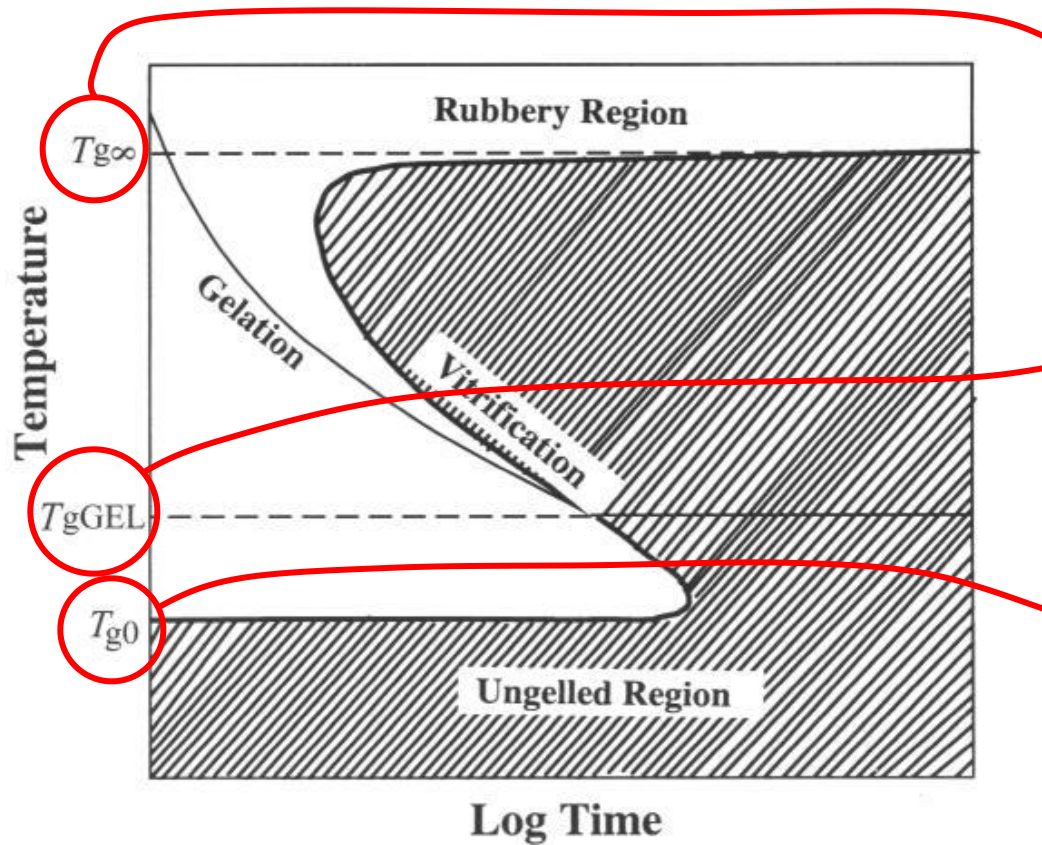


TTT diagram Depicts:

- Important temperatures
- Transitions between states

Kind of like a phase diagram for metals... *remember...*

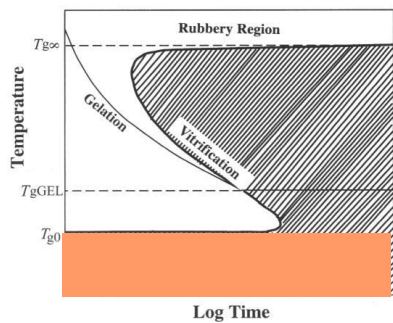
Note: *log time scale!*



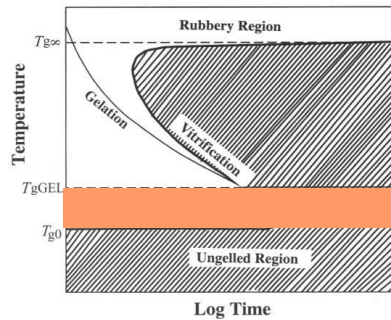
$T_{g\infty}$ is the glass transition temperature of the fully crosslinked polymer.

T_{gGEL} is the temperature at above which gelation occurs before vitrification.

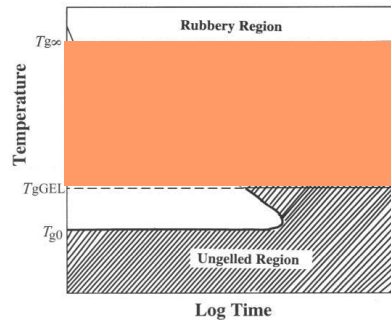
T_{g0} is the glass transition temperature of the unreacted components.



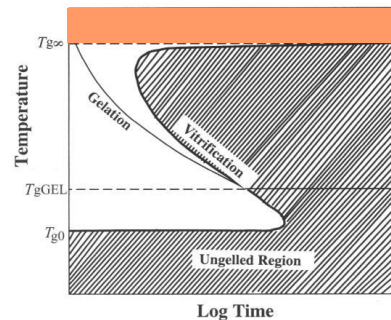
Below T_{g0} the catalysed solution will be a glassy solid. The crosslinking reaction can only occur **very** slowly, by diffusion (months or years). Thermosets used in preregs are stored below T_{g0} .



Between T_{g0} and T_{gGEL} , the catalysed solution is in a liquid state. Crosslinking occurs until vitrification, where a transition to a glassy solid is made. The reaction is very slow thereafter.

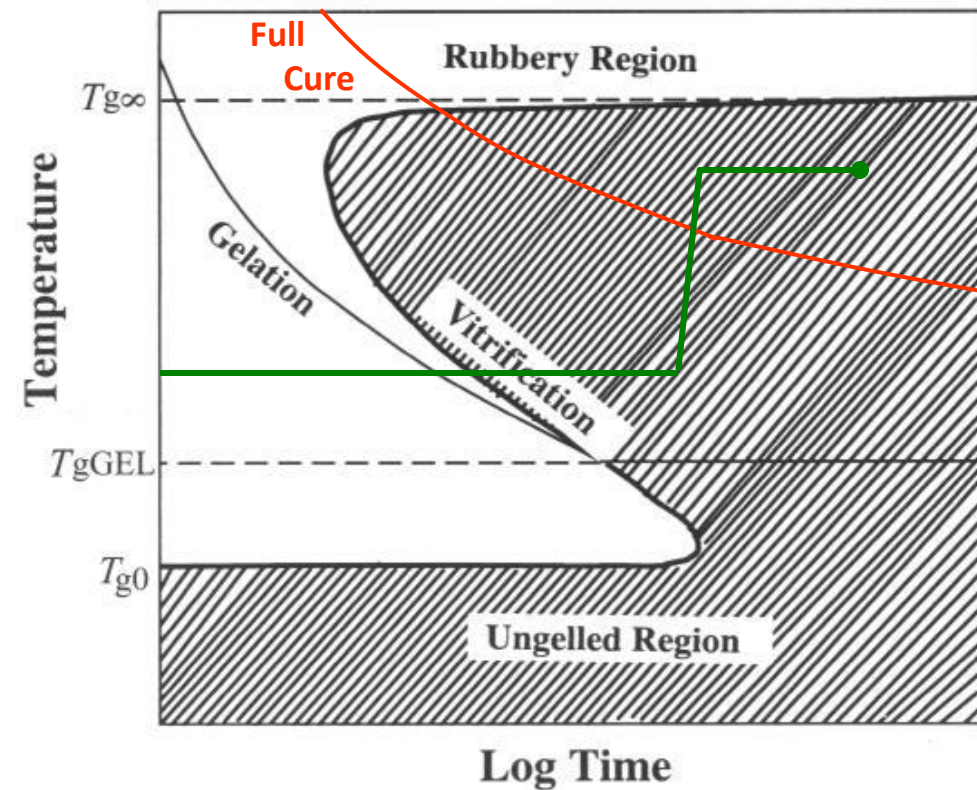


Between T_{gGEL} and $T_{g\infty}$, the liquid catalysed solution gels first, ending the possibility for flow. Crosslinking continues at a good rate until vitrification.



Above $T_{g\infty}$, the liquid catalysed solution gels, but will never vitrify. The polymer remains in a rubbery state throughout the curing process. Once temperature is brought below T_{g0} , vitrification occurs.

At vitrification the resin will not necessarily be 100% cured. Some amount of the unreacted components remain, and the reaction will continue very slowly from this point.



The **full cure line** on the TTT diagram denotes when the cross-linking operation is completed.

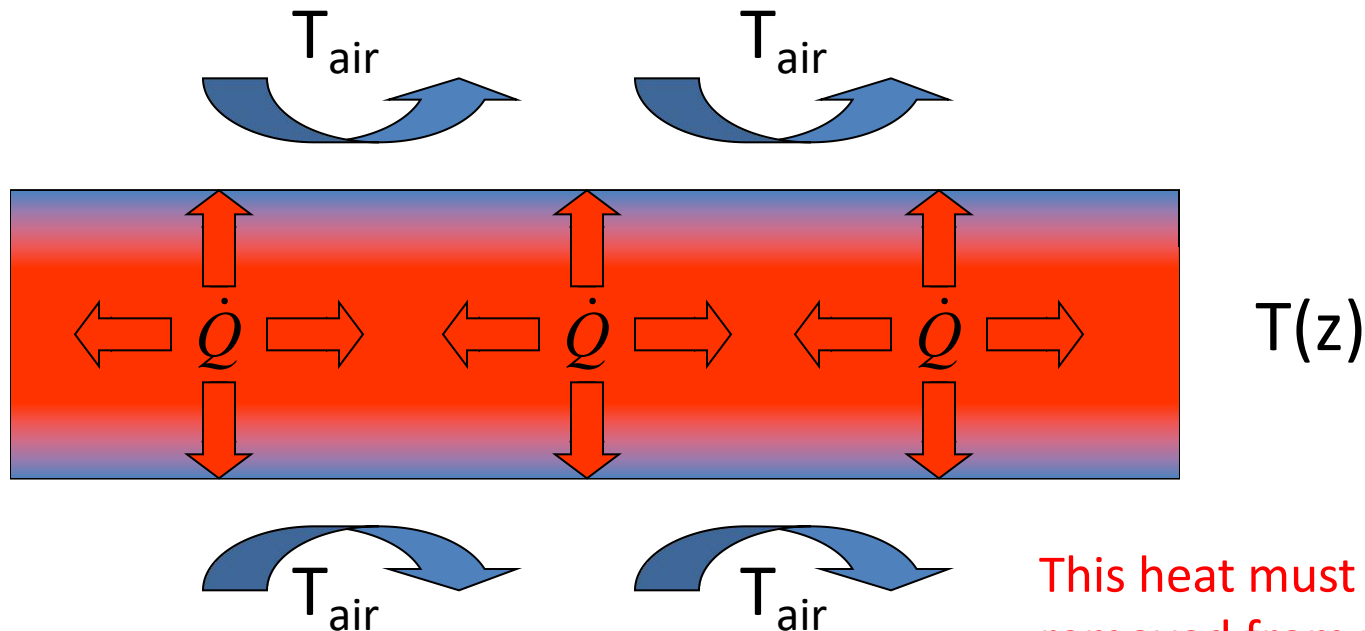
Some parts are **post-cured** at a higher temperature, for a significant time, to ensure full cure, and the best properties (long times though).

The Thermoset Cure Reaction

Thermoset cure reactions are **highly exothermic**, generating significant heat during the crosslinking process.

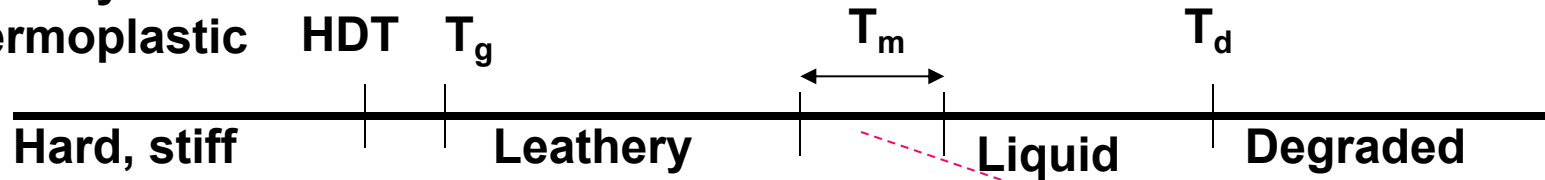
Not only are these reactions exothermic, but their **reaction rates** are also **affected strongly by the local temperature** of the resin.

Consider the “slab” of thermoset resin curing below:

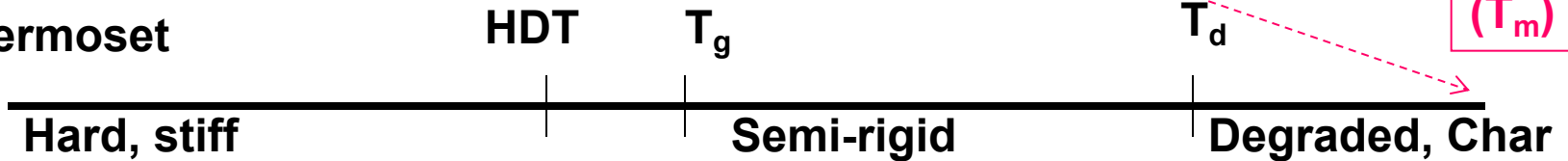


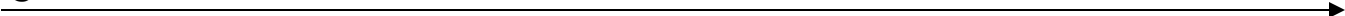
This heat must be removed from the part.

**Semi-crystalline
thermoplastic**



Thermoset

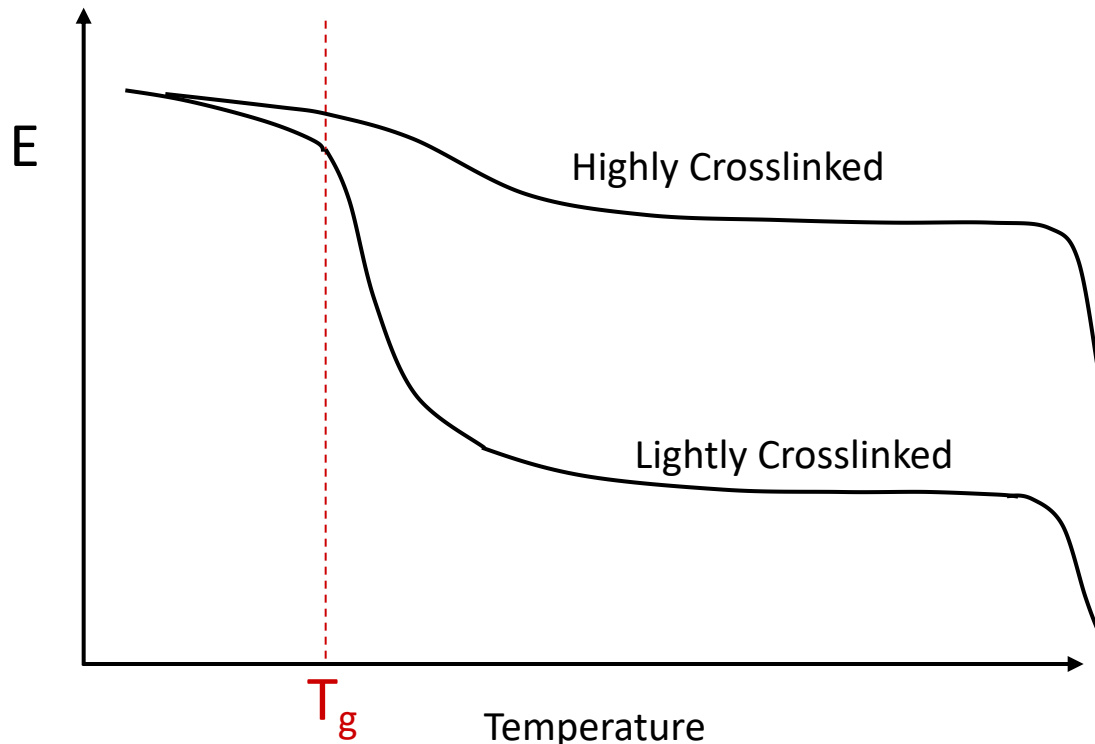


Temperature 

Solid Thermoset Properties

Due to crosslinking between polymer chains, thermosets are typically stiffer, **but** more brittle than thermoplastics.

Their resulting stiffness is a function of the degree of crosslinking, and the application temperature.

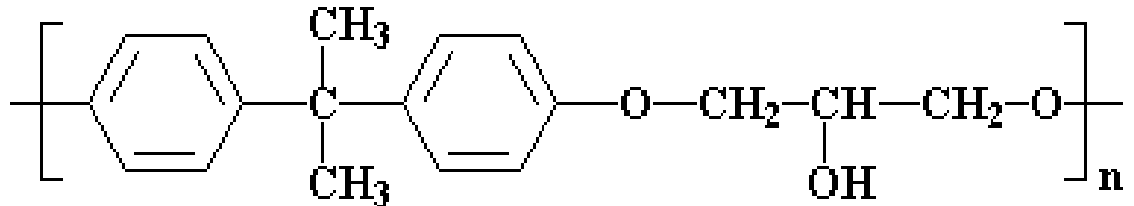


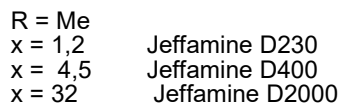
While a **crosslinked thermoset will not melt,**

degradation of the polymer will occur above a certain temperature.

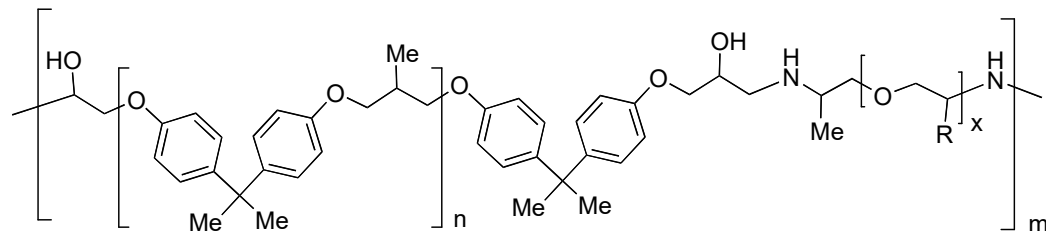
Epoxies

- **Epoxy resin** is made from the 2-part kits.
- It's the basis of composites like fiberglass, carbon fiber composites
- Apart from an excellent glue, it is an important molding compound for rapid prototyping.
- Tensile strength 60 MPa
- Modulus 2.6 GPa
- Chemical and corrosion resistant
- Low shrinkage
- Cures with amines, alcohols and carboxylic acids



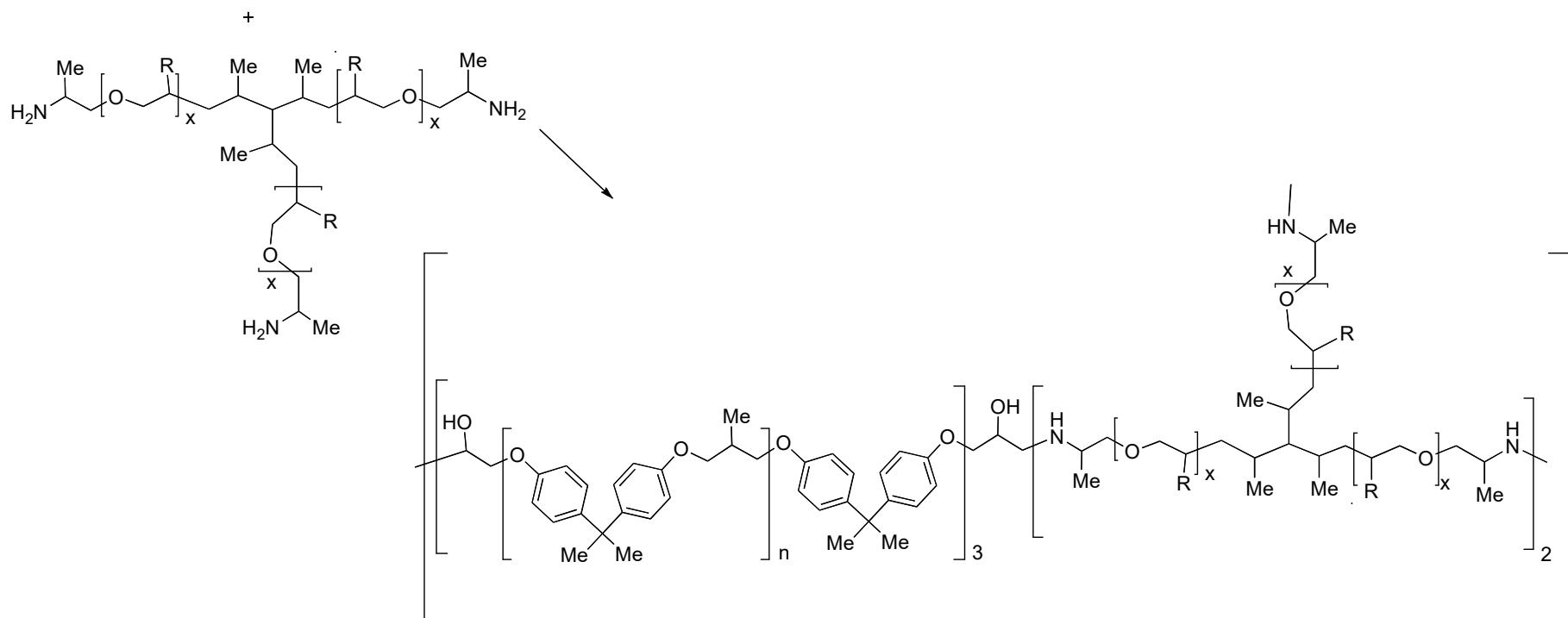
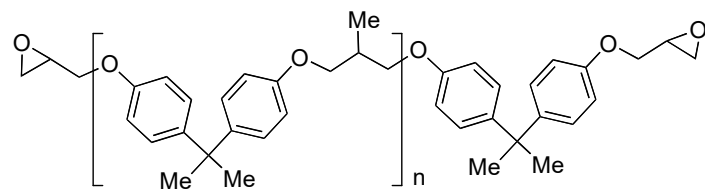


catalyst

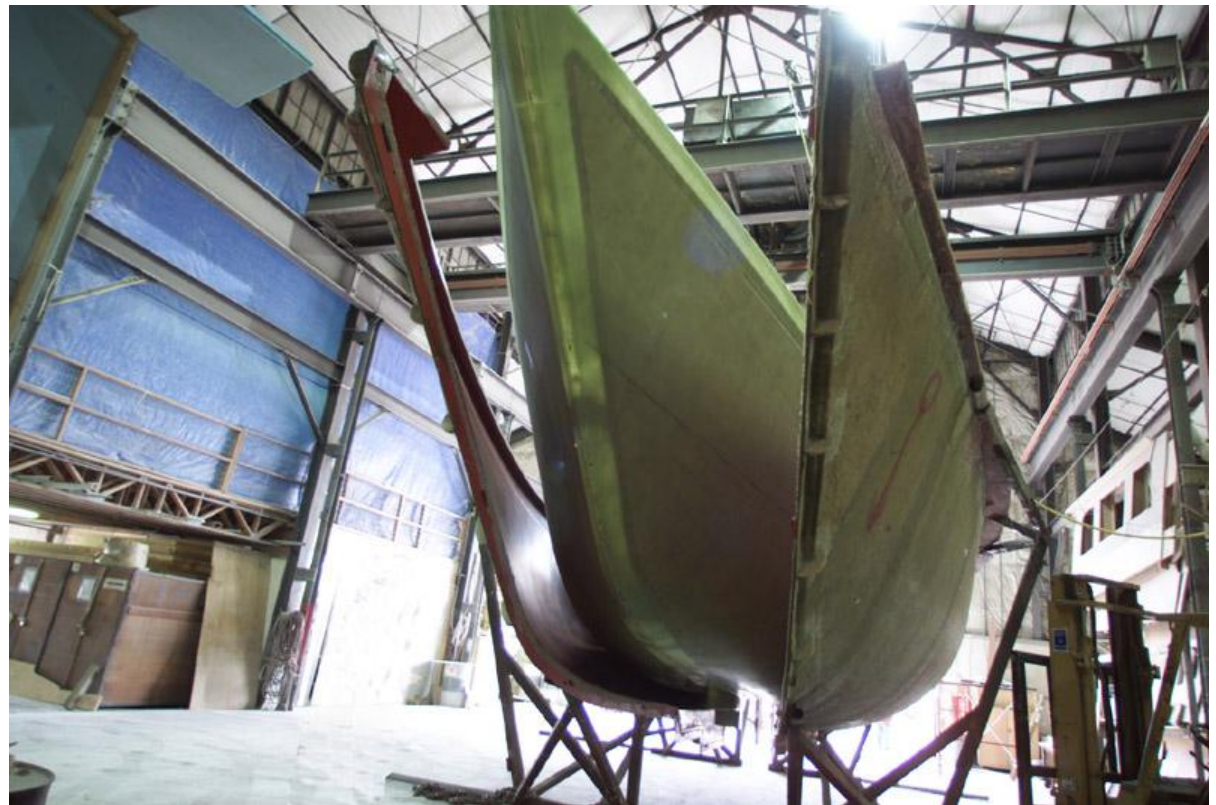


Linear Cured Epoxy

Insoluble Epoxies: Branched Polyamines



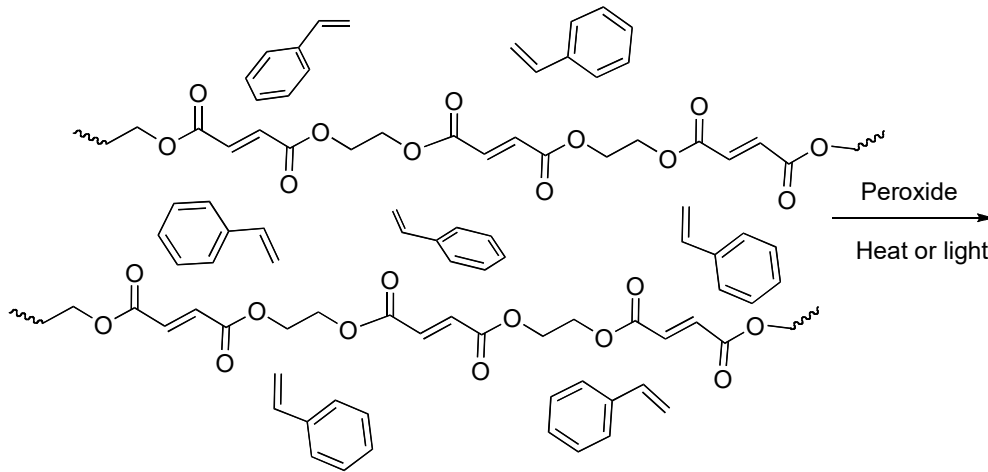
Epoxies



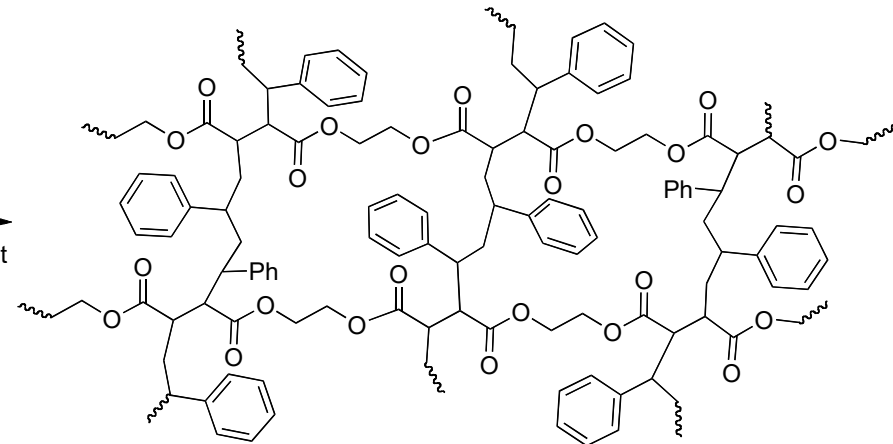
Polyester Thermosets (TS) or Unsaturated Polyesters (UP)

- Largest group of thermosets
- Most like to be reinforced with fiberglass

Solution of Polyester and styrene

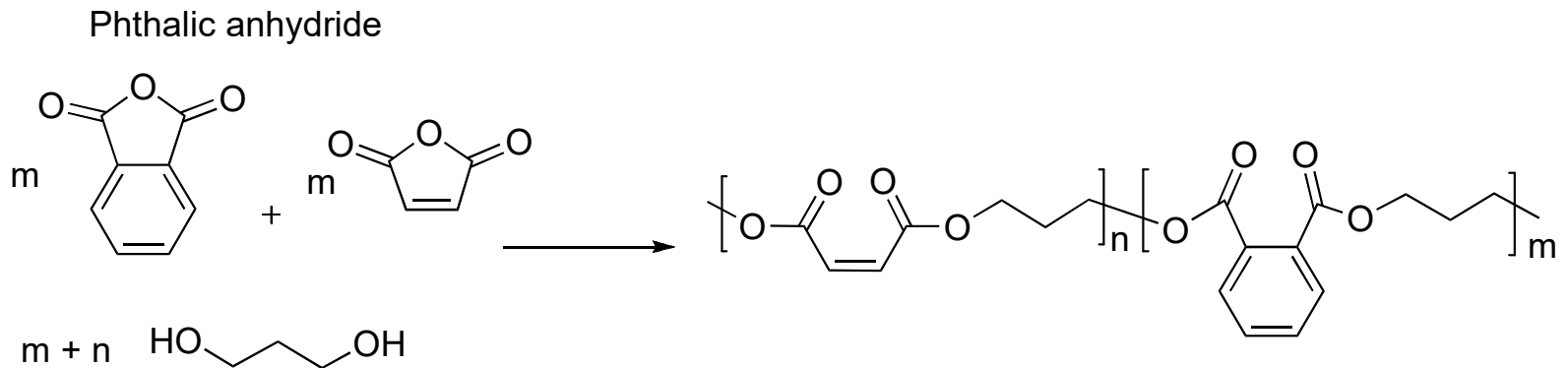
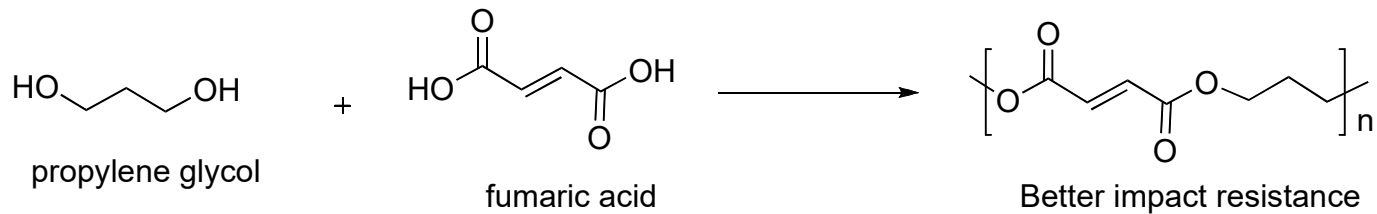
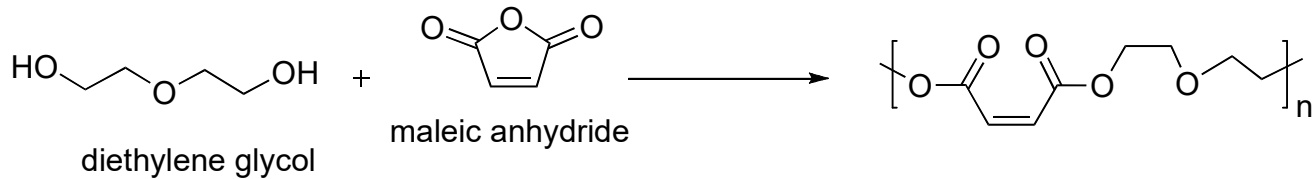


Crosslinked thermoset



“Casting Resin”

Unsaturated Polysters

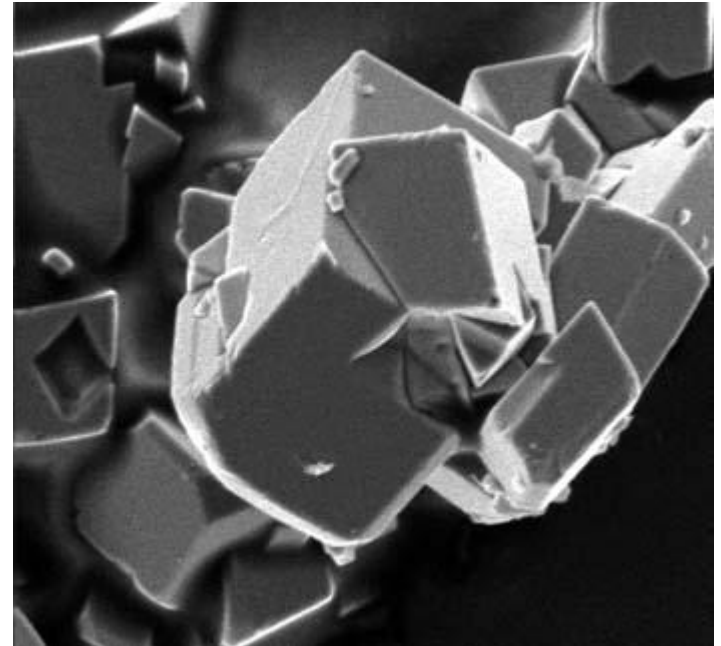


Reduced cross-link density: lower modulus , less brittle

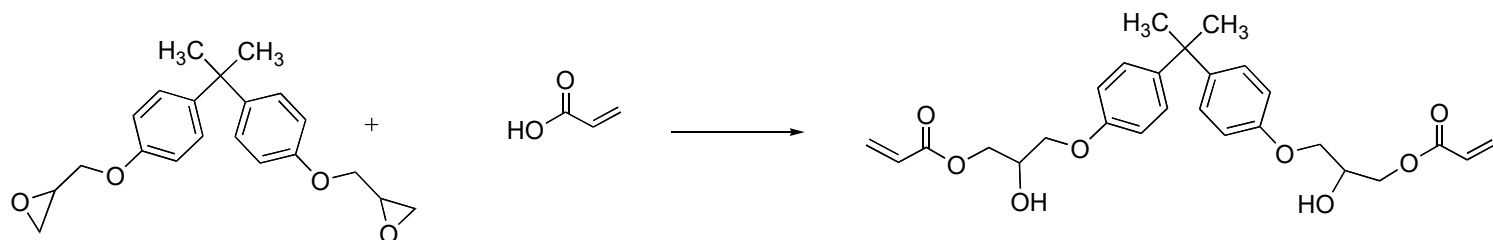
Additives to Thermoset Polyesters



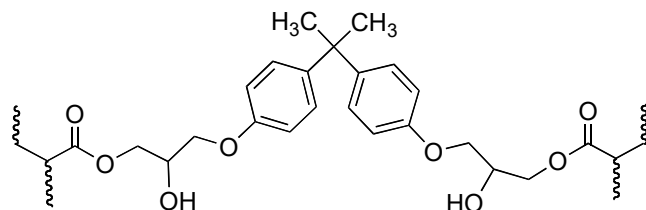
- **Fillers like Calcium Carbonate**
- **Tougheners like rubbers**
- **Antioxidants and UV stabilizers**
- **Reinforcements**



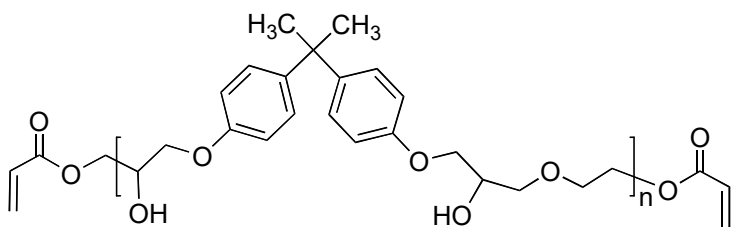
Vinyl Esters (VE)



peroxide
Heat

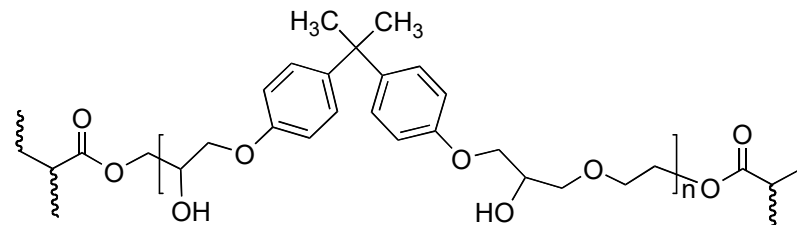


Thermoset



Thermoset

peroxide
Heat



Thermoset

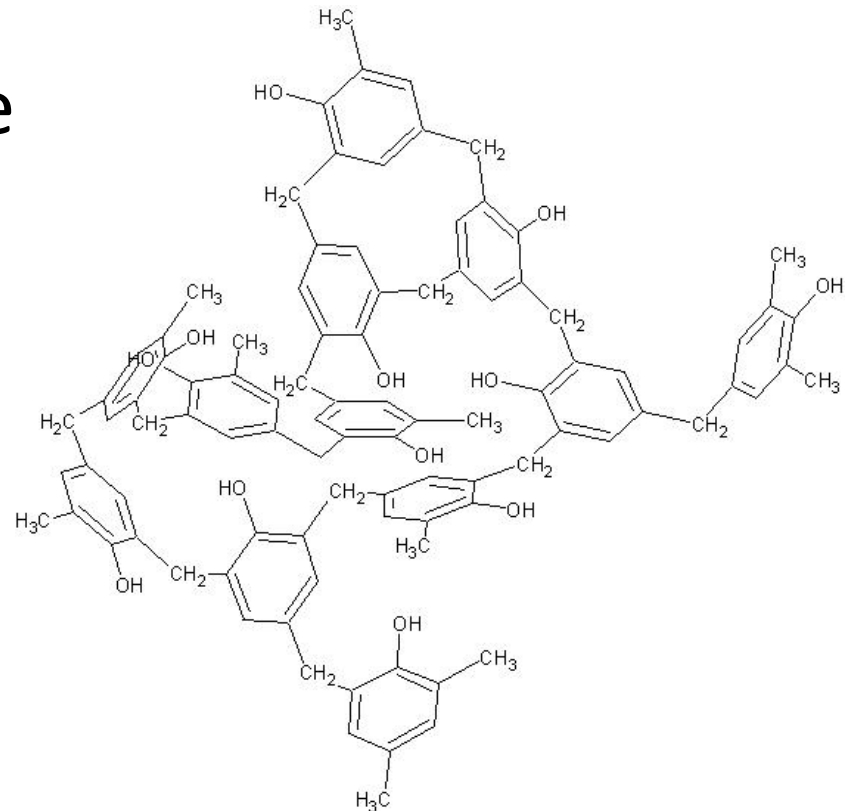
Intermediate between polyesters & epoxies in performance and cost

Vinyl Esters (VE)

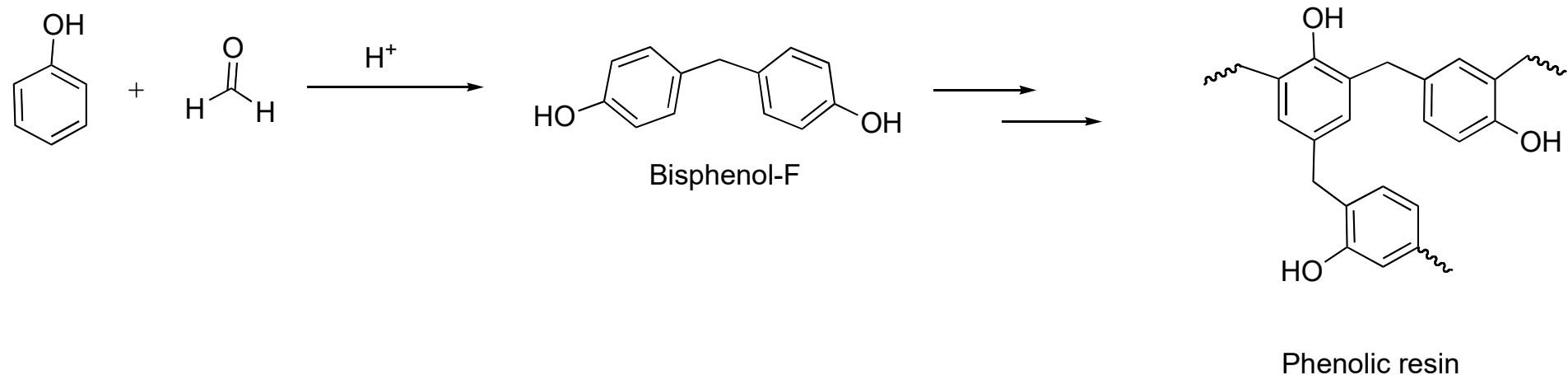


Formaldehyde Resins

- Phenolic
- Urea formaldehyde
- Melamine formaldehyde

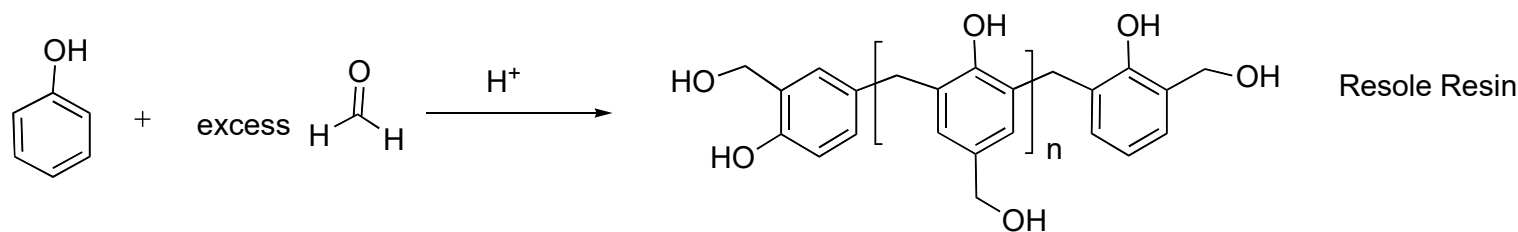
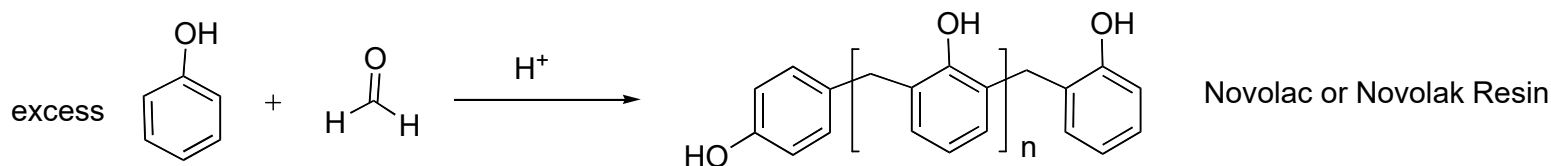


Phenol-Formaldehyde Resins



Residual formaldehyde in cross-linked matrix

Phenol-Formaldehyde Resins



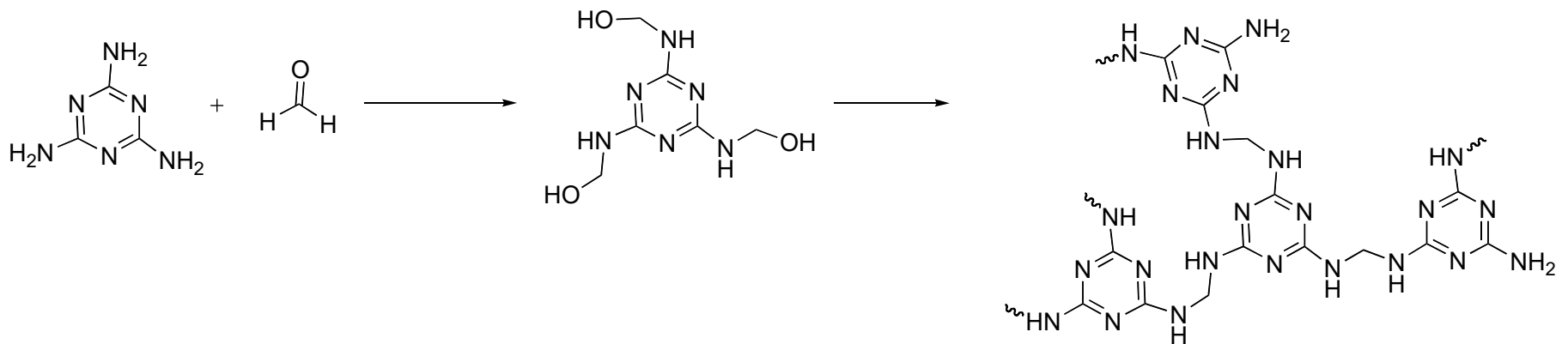
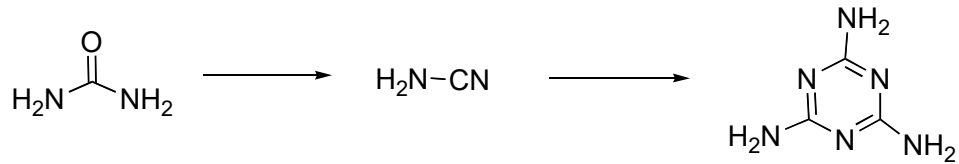
Novolacs are widely used photoresists
Both of these are reactive thermosets

Thermoset Types

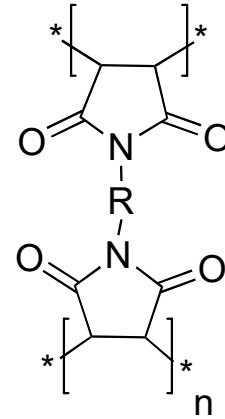
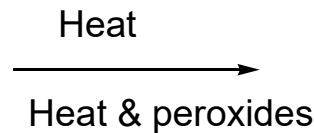
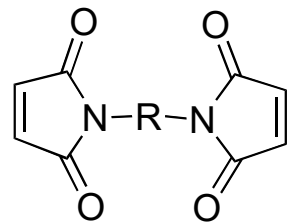
- Phenolics



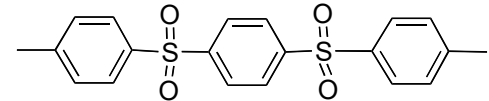
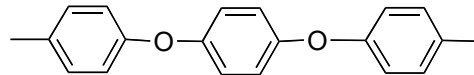
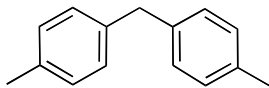
Melamines



Polybismaleimides



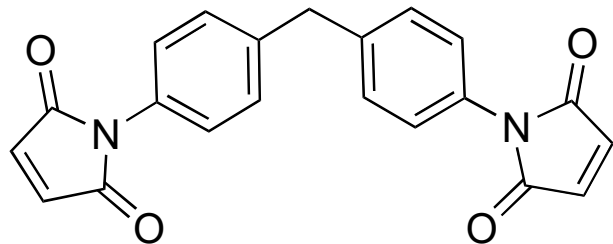
R =



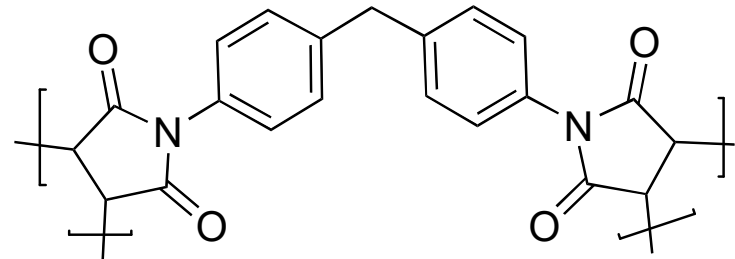
Aliphatics, mixed bridging groups

Printed wiring boards, carbon fiber composites for aerospace
Brittle but can be toughened with chain.

Polybismaleimides



Mp 155 ° C



T_g

> 500 ° C

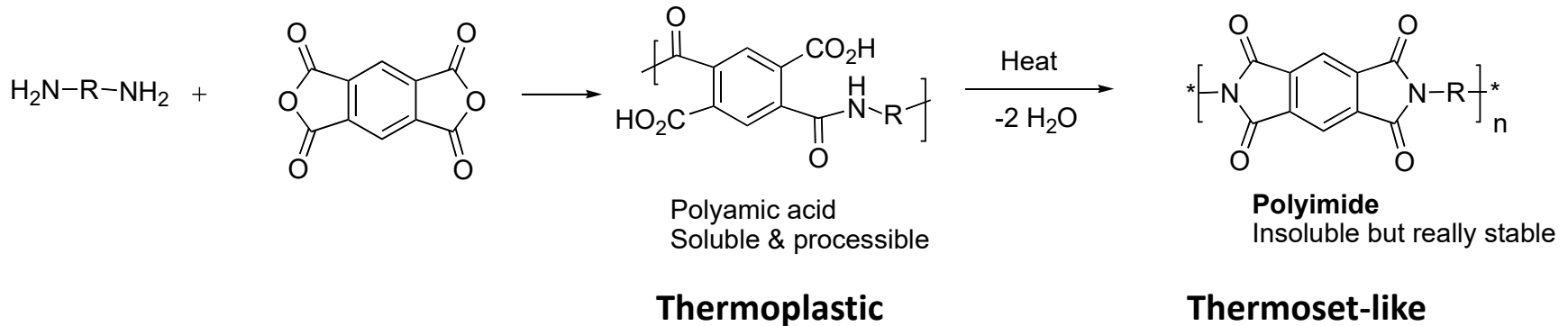
Tensile Strength

41-83 MPa

Tensile Modulus

4-5 GPa

Polyimides



High operating temperatures (up to 500°C)

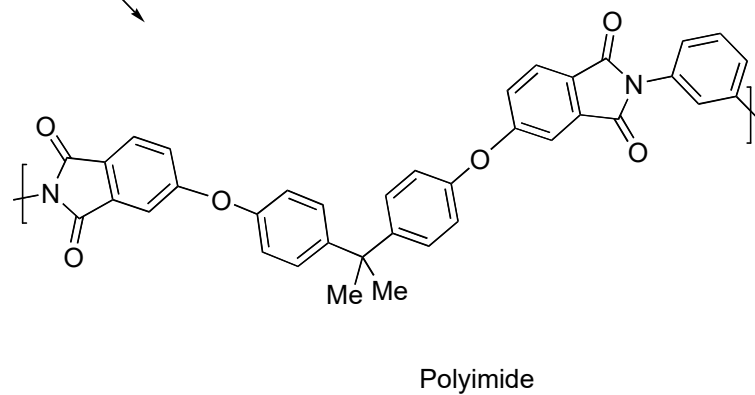
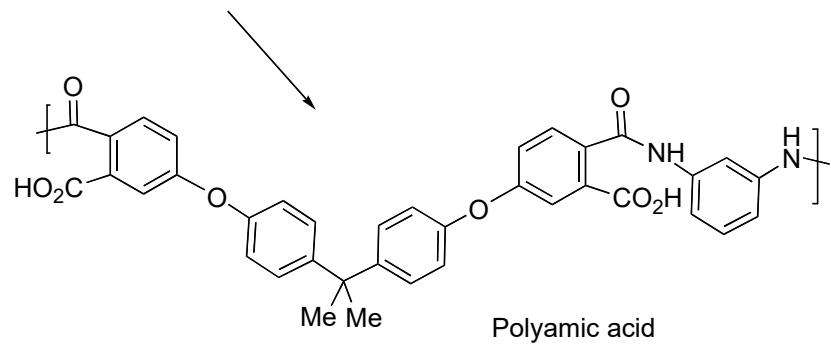
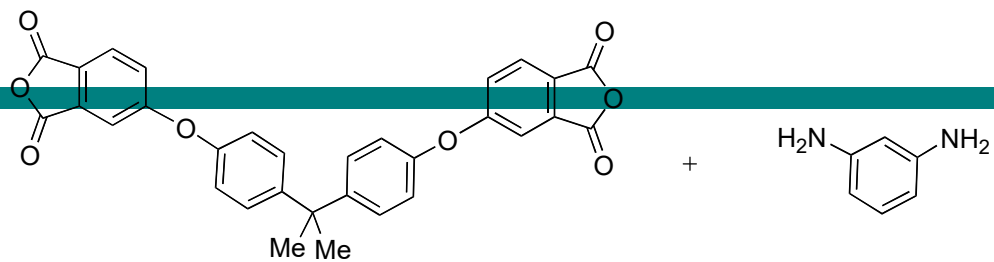
High tensile strength & modulus

Low creep and outgassing

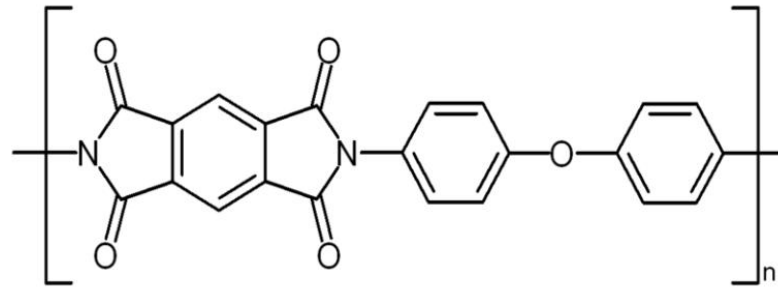
Flame resistant

Solvent resistant

Composites, high temperature adhesives, wire insulation for extreme environments



Polyimides:Kapton

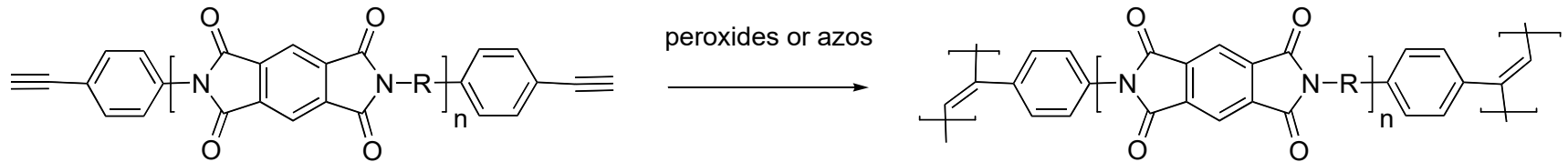
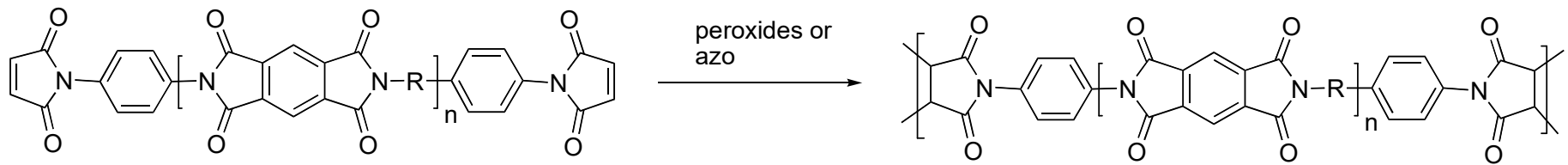


Operating temperature range: -269°C to 400°C

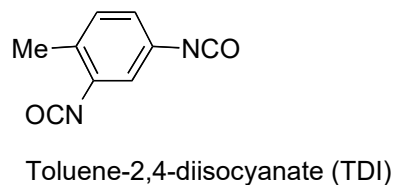
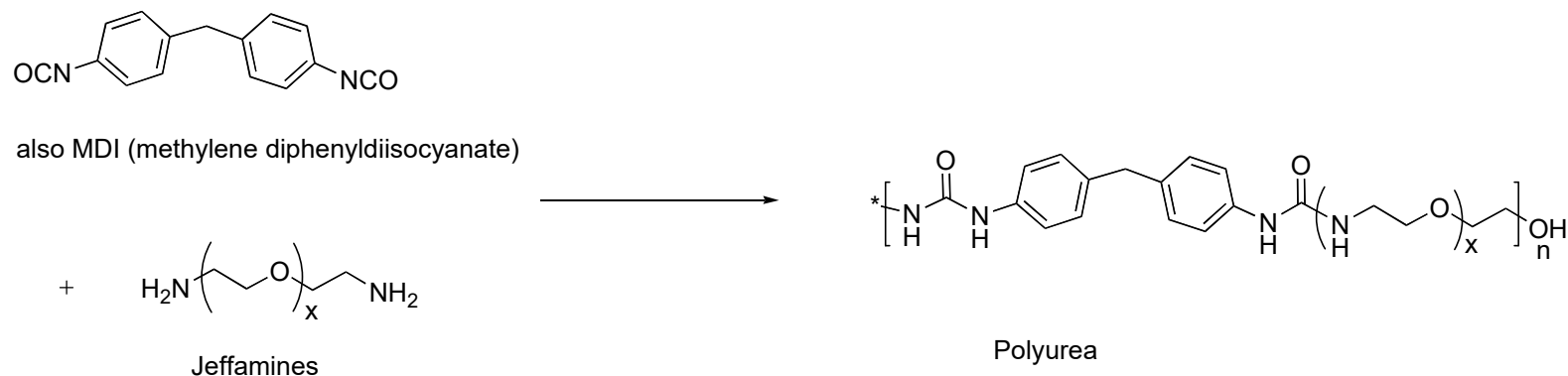
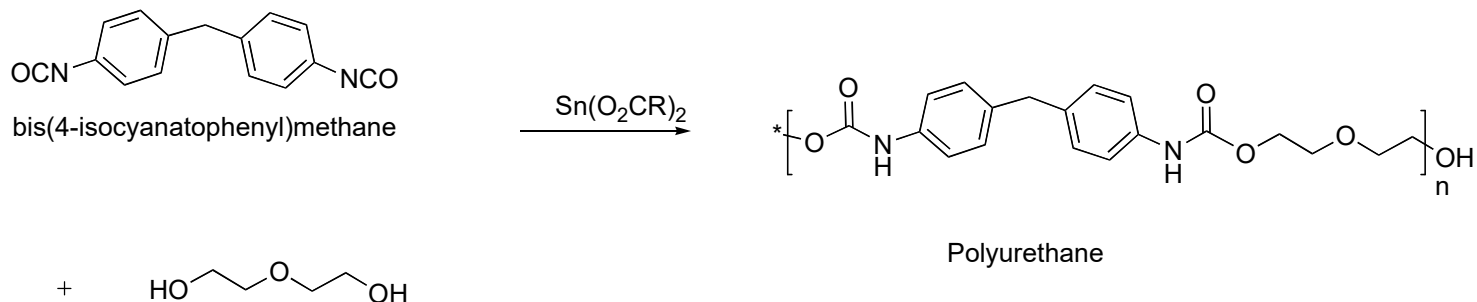
- Composites, space suits, dielectric material for printed wiring boards
- Poor resistance to mechanical wear: wiring in aircraft shorted out due to Kapton failure

Kapton(DuPont)

Polyimide Thermosets

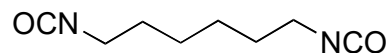


Polyurethanes



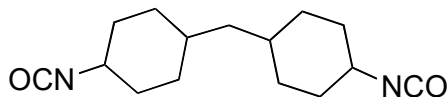
Soft or thermoplastic elastomers

More Flexible Polyurethanes



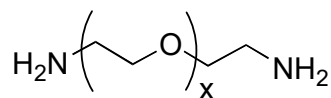
hexamethylene diisocyanate (HMI)

of

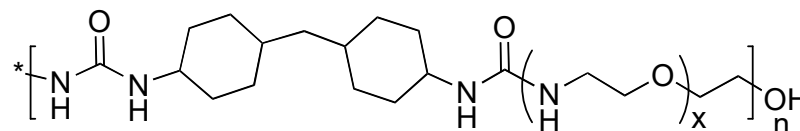
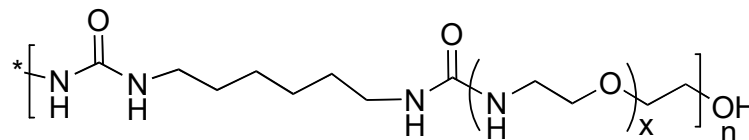
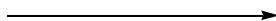


H12 MDI

+

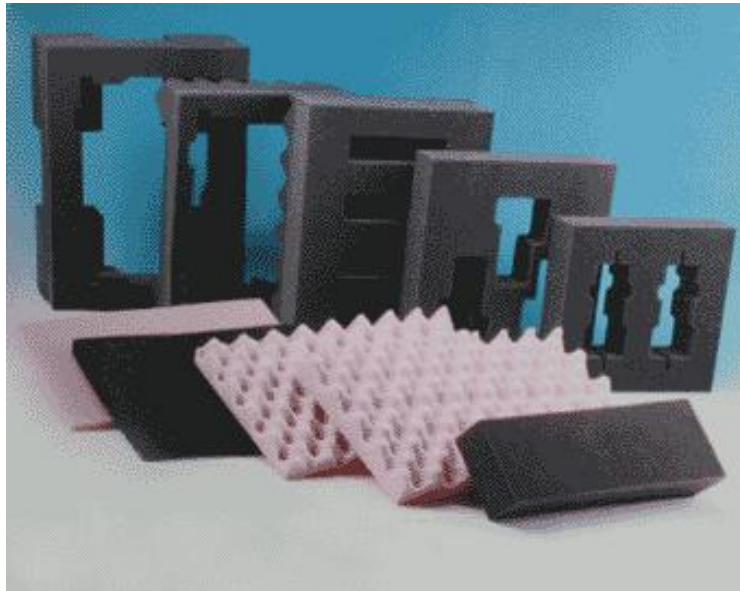


Jeffamines

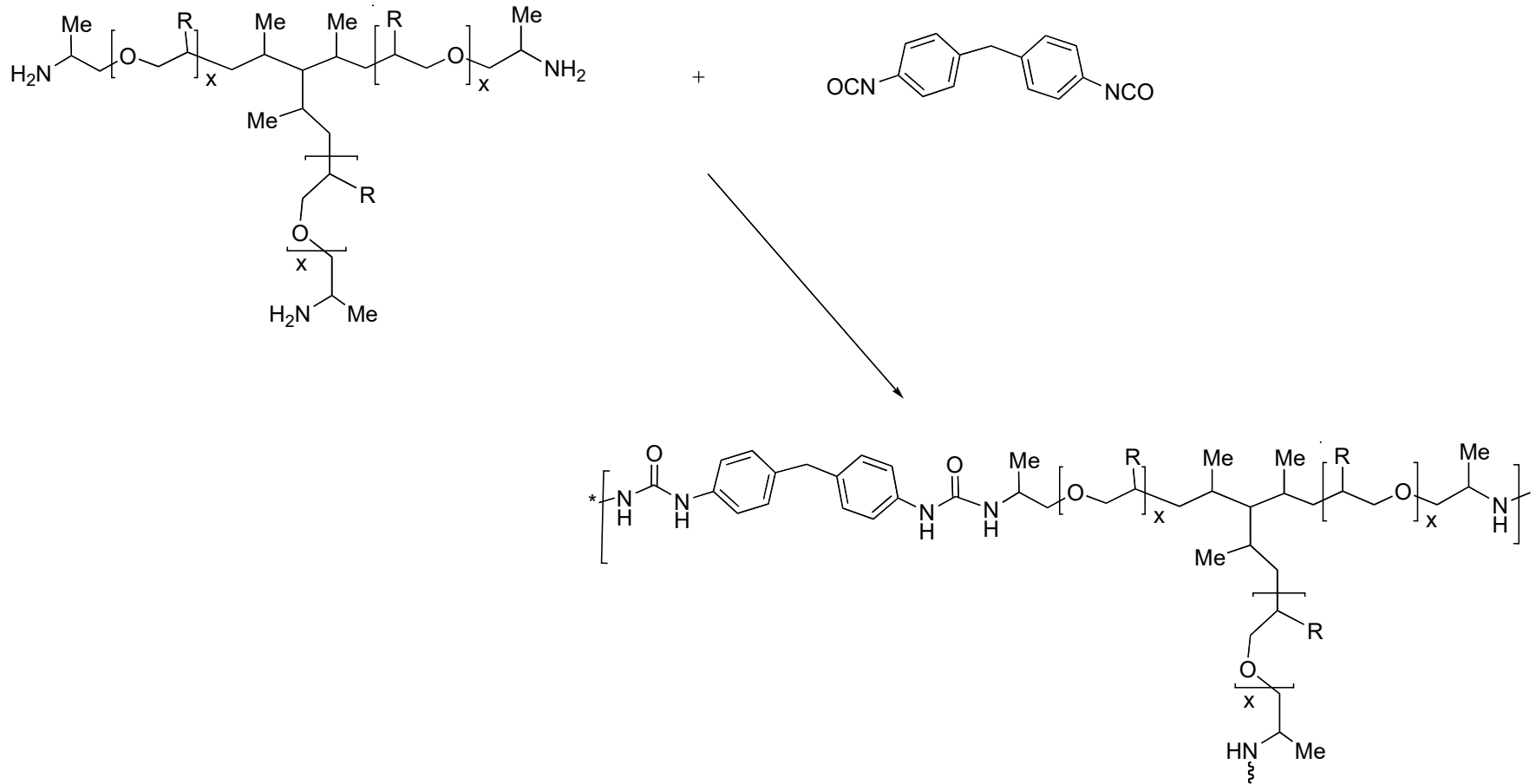


More flexible Polyureas

Polyurethanes

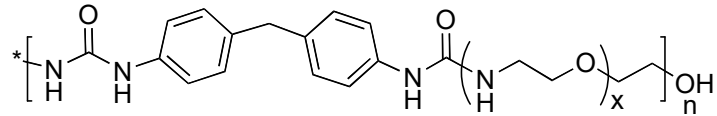
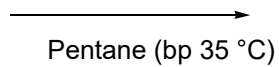
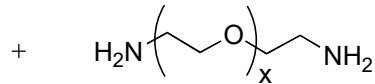
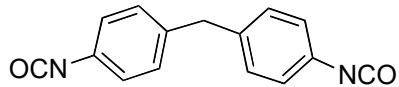


Polyurethane or Polyurea Thermosets



Rigid-high modulus

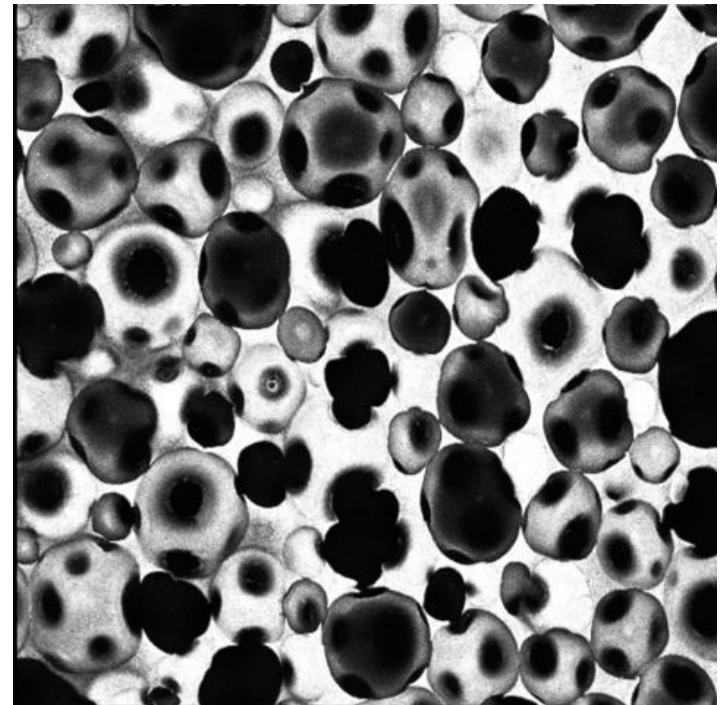
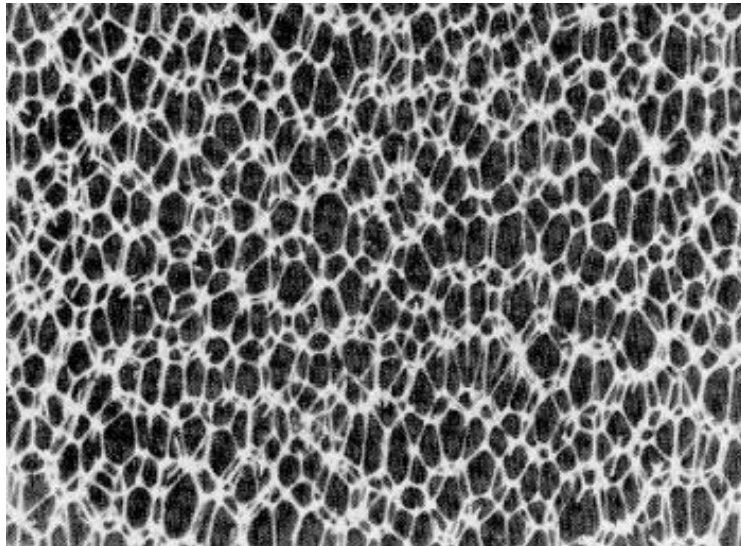
Polyurethane Foams



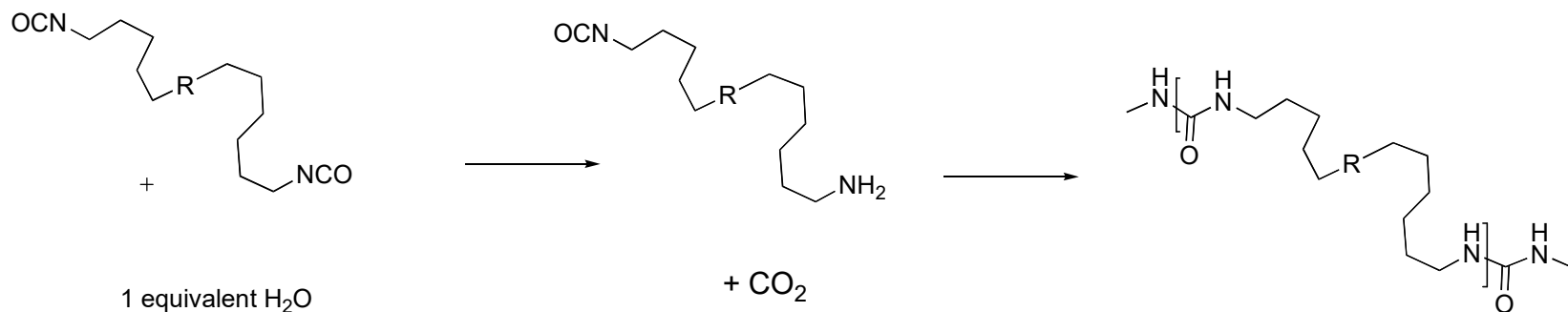
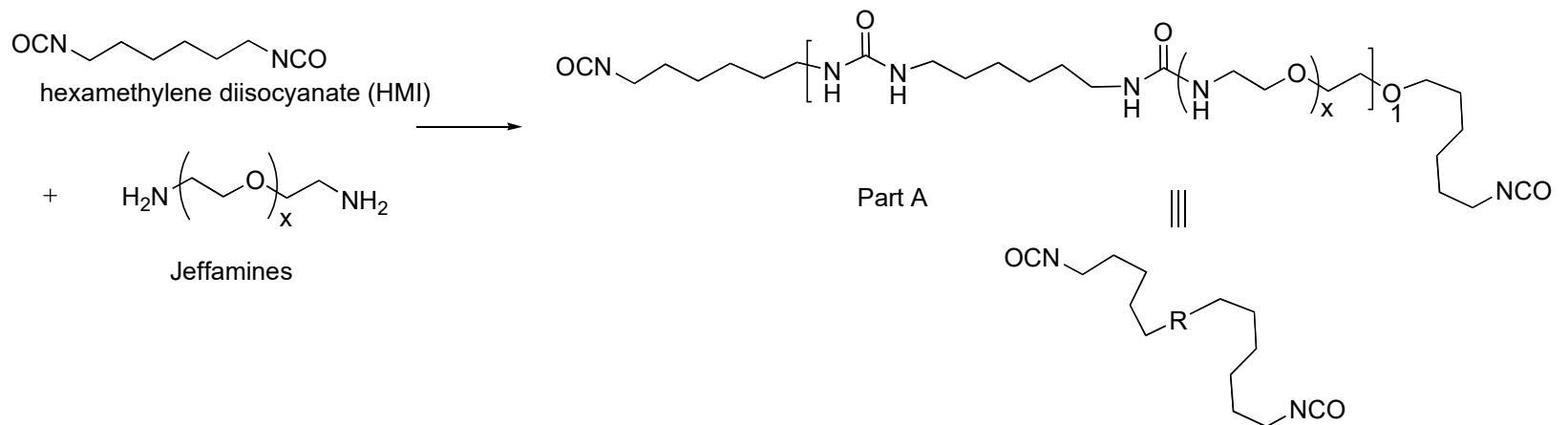
Soft foam

Polyurea Foam

Solvent blown foam



Green Polyurethane Foams



Carbon dioxide is generated as blowing agent for foam

A variety of solid properties are relevant to manufacturing;

	Density ρ kg/m ³	Young's Mod E GPa	Tensile Strength σ MPa	Strain to Failure ϵ %
Polyester	1100-1230	3.1 – 4.6	50 – 75	1.0 – 6.5
Vinylester	1120-1130	3.1 – 3.3	70 – 81	3.0 – 8.0
Epoxy	1100-1200	2.6 – 3.8	60 – 85	1.5 – 8.0
Phenolic	1000-1250	3.0 – 4.0	60 – 80	1.8
PUR	1200	0.7	30 – 40	400-450
BMI	1200-1320	3.2 – 5.0	48 – 110	1.5 – 3.3
PI	1430-1890	3.1 – 4.9	100 – 110	1.5 – 3.0
Aerospace Al	2800	72	>540	
Carbon Steel	7790	205	640	

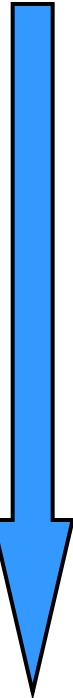
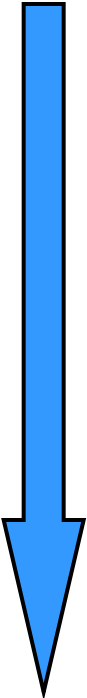
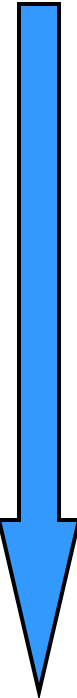
* At room temperature

** Last two materials provided for reference

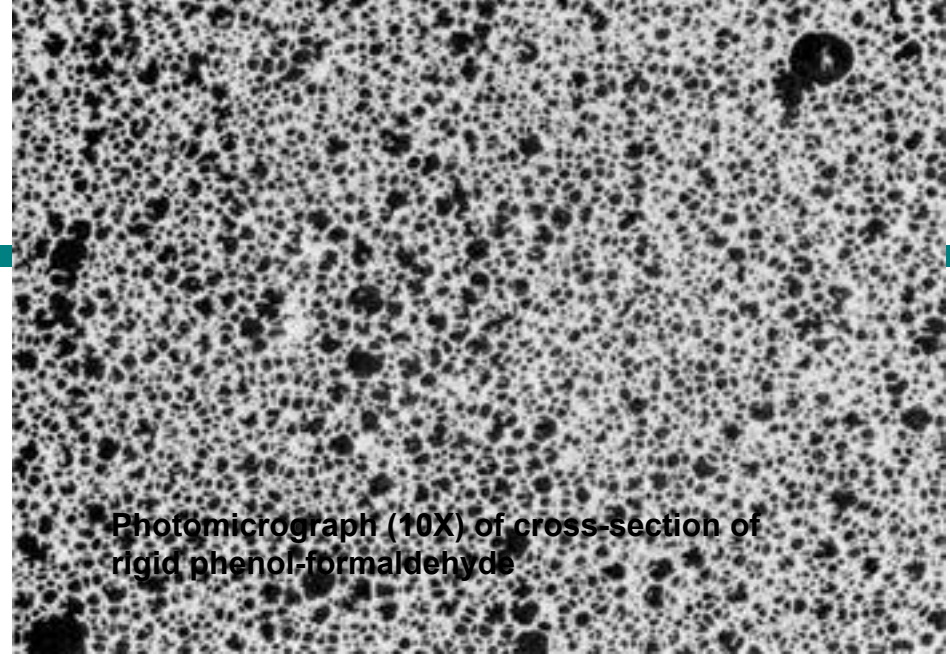
Common Shaping Processes for Thermosets

	COMPRESSION MOLDING	TRANSFER MOLDING	INJECTION MOLDING	ROTATIONAL MOLDING	THERMOFORMING	REACTION INJECTION MOLDING	CASTING	FOAM MOLDING	REINFORCED PLASTICS MOLDING	LAMINATING
Alkyd	■	■	■				●		●	
Allyl					●		●		●	■
Epoxy				●		■	●	●	●	●
Maleimide	■	●	●					●	■	■
Phenolic	●	●	■				●	■	●	●
Polyester	■					●	■	●	●	
Polyurethane						■				
Silicone							●	●	■	
Urea	●	●	■					●		

Common Thermosets applied to Composites

Resin Type	Performance	Processing	Cost
• Polyester • Vinyl Ester • Polyurethane (PUR) • Epoxy • Bismaleimide (BMI) • Phenolic • Polyimide (PI).	LOW  HIGH	EASY  DIFFICULT	LOW  HIGH

Polymeric Foams



Photomicrograph (10X) of cross-section of rigid phenol-formaldehyde

- Polymers can be combined with a gas
 - Forms voids or cells in the polymer causing the polymer to be very light
 - Referred to as cellular, blown, expanded polymer, foam
 - Elastomeric foam- matrix (polymer) is an elastomer or rubber
 - Flexible foam- soft plastic matrix, e.g., plasticized PVC (PPVC), LDPE, PU
 - Rigid foams- PS, unsaturated polyesters, phenolics, urethanes (PU)
 - Type of polymer matrix, thermoplastic or thermoset can form basis for classification
 - Amount of gas added reflects the resulting density
 - Light foams: density = 0.01 to 0.10 g/cc (1 to 6 lb/ft³)
 - Dense foams: density = 0.4 to 0.6 g/cc (25 to 40 lb/ft³)
 - Note: water = 1g/cc or 62.3 lb/ft³

Polymeric Foams

Mechanisms for the formation of cellular structure

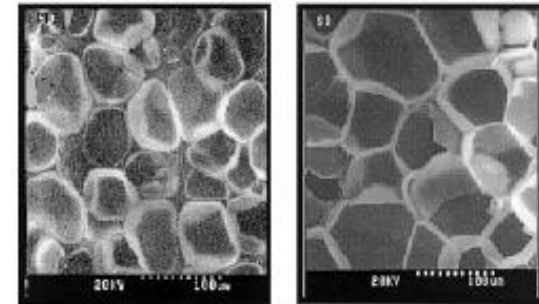
–Aeration or frothing: mechanical agitation is used to incorporate air into liquid resin system (latex, reactive urethane)

–Physical blowing agent:

- Add N_2 gas into solution or to liquid melt which comes out of solution when pressure is released and forms cells.
- Add liquids at room temperature and have low boiling point. The liquids vaporize upon heating or by chemical reaction heat.

Polystyrene: PS or expanded polystyrene foam (EPS)

- Made from expandable polystyrene beads which are small spheres of polystyrene (diameter of 0.3 – 2.3 mm) containing 3-7% pentane as physical blowing agent
 - Bulk density of beads (with air spaces) is 0.7 g/cc.
- Manufacturing
 - Beads are pre-expanded with the use of a steam chamber to a bulk density of 0.02-0.05 g/cc.
 - Beads are cooled and reached equilibrium with air penetrating the cells.
 - Placed back in steam chamber and molded into final foamed shape.
 - » Forms basic cellular structure is closed cell type
 - Large blocks are molded which are cut into insulating boards or molded into custom products
 - » Cups, insulating containers, protective elements
 - Extrusion process can be used with blowing agent
 - » Meat trays, egg cartons

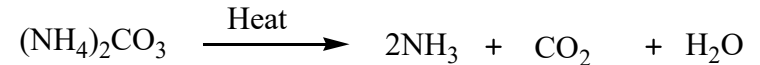


Micrographs of the specimens produced at foaming temperatures (a) 105°C and (b)

Chemical Blowing Agents

Compounds that decompose under heat and liberate large amounts of inert gas,

- N_2 , CO_2 , CO, water, ammonia, H_2 , etc.



- Activators can sometimes be added to allow lower decomposition temperature and release more gas at a lower temperature.

- Early blowing agents were

- Sodium bicarbonate, which liberates CO_2

- Other carbonates and nitrates liberate hydrogen or nitrogen.

- Hydrogen can be generated in large quantities, but diffuses away quickly

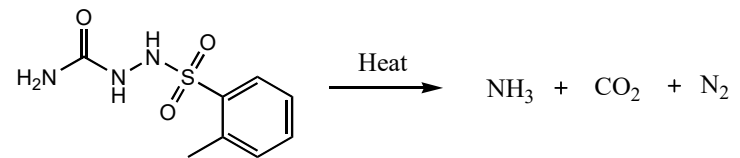
- Organic compounds can be used for some high temperature thermoplastics

- Toluene sulfonyl hydrazine

- azodicarbonamide

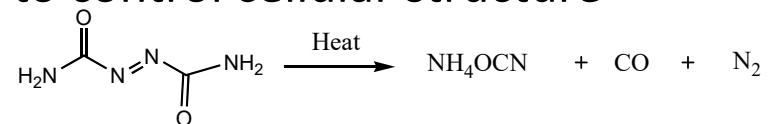
- Toluene sulfonyl semicarbazide

- Phenyl tetrazole



- Can be in finely divided solid form to create cellular structure

- Nucleating agents and surfactants are used to control cellular structure



azodicarbonamide